

THE ELEMENTARY AND PRINCIPLES OF
FINANCE

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TO THE READER

This book is written for readers who are meeting the subject for the first time. It assumes no earlier knowledge. Every new word is explained when it first appears, and the book moves slowly wherever an idea is new or difficult.

Beside most paragraphs you will find a note in the narrow column at the edge of the page. A note in italics gives the main point of the paragraph in one line. Read these notes first to see the shape of a chapter, and again when you review.

A word in **bold type** is a new term. It is shown in bold once, where it is first explained, and its meaning is repeated in the narrow column. After that the word appears in ordinary type. Italics are used for emphasis, for foreign words and for the titles of books.¹

Where an everyday word has a special meaning here, a note marked "Word in use" says so. Where an idea is both new and hard, a note reads "Go slowly here", and the text slows down with it.

HOW A CHAPTER RUNS

A chapter opens with a puzzle, then builds its ideas one section at a time. Each section ends with a short recap. The chapter closes with a summary, a problem to think through, the common mistakes people make, a true account from history, and review questions. Questions are numbered, such as 7.1, and their answers are collected at the back of the book.

When a chapter points to another part of the book, it names the chapter and the section, for example Chapter 3, section 3.2. Use the Contents to find it. At the back you will find a glossary of every term, the answers to the questions, a list of further reading, and an index.

*This book teaches *The Elementary and Principles of Finance* one idea at a time, from the very beginning.*

The narrow column beside the text holds short notes.

Key term. A new word, shown in bold where it first appears, with its meaning in one line.

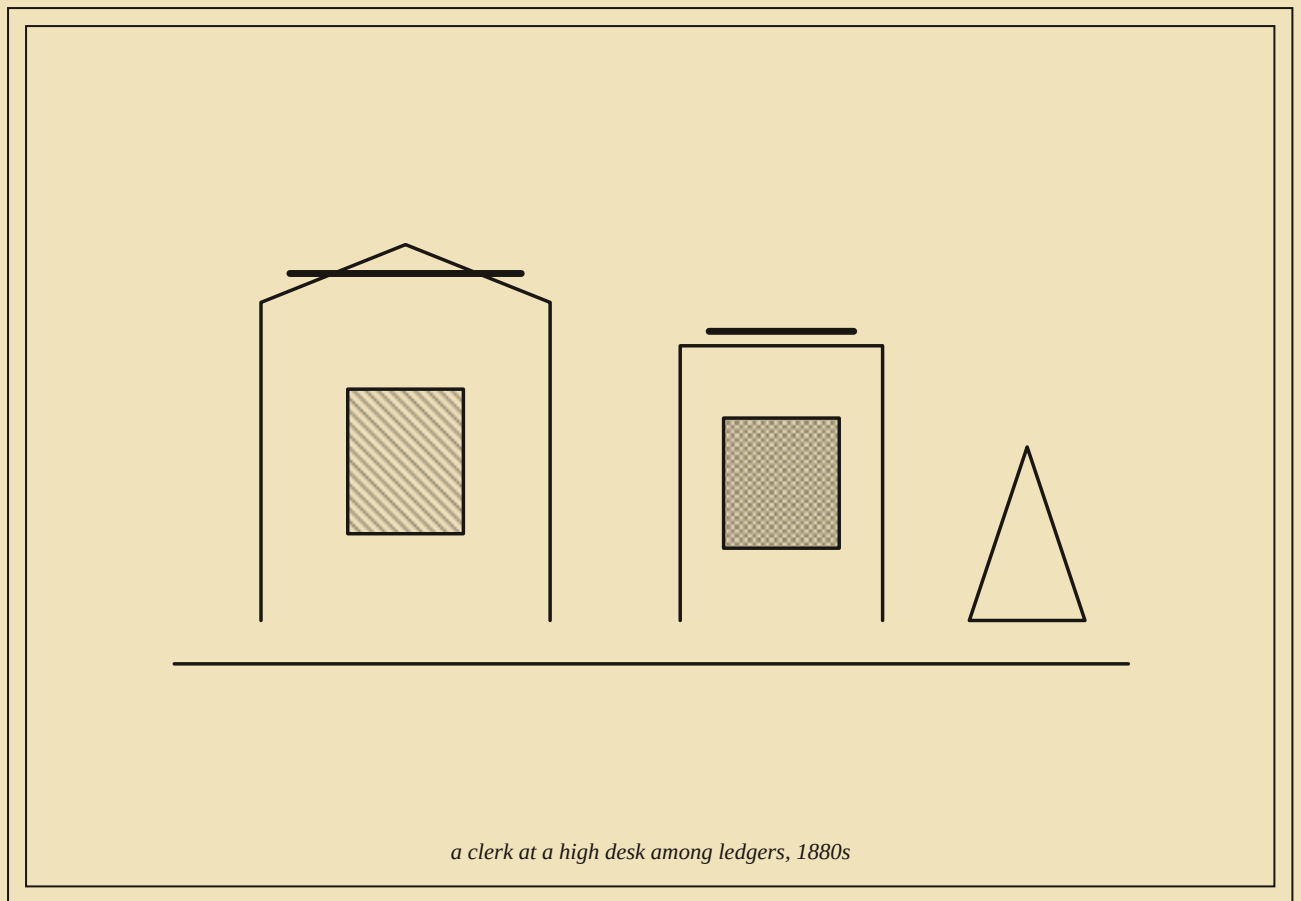
A bold word is a new term, and its meaning waits beside it.

Some everyday words have a second, special meaning in this subject.

Each chapter has the same shape, so you always know where you are.

Chapters refer to one another by name, never by page.

¹ A footnote gives a side point that would interrupt the main argument. Footnotes are numbered within each chapter and appear at the foot of the page.



CHAPTER I

Scarcity, Choice and the Economic Problem

What does this chapter let us explain that the earlier ideas could not explain alone?



Show why every society and every shop must choose, and give the reader the first working words of economics. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Show why every society and every shop must choose, and give the reader the first working words of economics. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet; two paid assistants. Record of 12 January 2026: restock fund Tk 400,000 in cash from family savings; shelf space allows one large restock only. Option A, textbooks: purchase cost Tk 400,000; expected sales Tk 560,000; expected sales less purchase cost Tk 160,000. Option B, stationery: purchase cost Tk 400,000; expected sales Tk 540,000; expected sales less purchase cost Tk 140,000. Decision recorded: option A. Opportunity cost of that decision: Tk 140,000. The three problems in shop terms: what to stock (textbooks), how to run the shop (family at the counter plus two assistants), for whom (university students first, schools second). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

1.1 | WANTS, RESOURCES AND SCARCITY | NEW IDEA | 775

This section asks what wants, resources and scarcity means before applying it.

Economics. the study of how people, firms and societies choose among uses of limited resources to meet wants

Economics is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Economics** means the study of how people, firms and societies choose among uses of limited resources to meet wants. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wants, resources and scarcity, before moving to the next step.

The term **Want** means something a person would like to have, whether or not they can pay for it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wants, resources and scarcity, before moving to the next step.

The term **Resource** means anything used to make goods and services: land, labour, capital and enterprise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wants, resources and scarcity, before moving to the next step.

The term **Factor of production** means one of the four kinds of resource used in making things: land, labour, capital, enterprise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wants, resources and scarcity, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in wants, resources and scarcity to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

Want. something a person would like to have, whether or not they can pay for it

Want is the working idea for this part of the chapter.

Resource. anything used to make goods and services: land, labour, capital and enterprise

Resource is the working idea for this part of the chapter.

Factor of production. one of the four kinds of resource used in making things: land, labour, capital, enterprise

Factor of production is the working idea for this part of the chapter.

Wants, Resources and Scarcity is understood by separating the question into small steps.

Wants, Resources and Scarcity is understood by separating the question into small steps.

Wants, Resources and Scarcity is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

1.2 | CHOICE AND OPPORTUNITY COST | NEW IDEA | 550

This section asks what choice and opportunity cost means before applying it.

Good. a physical thing people buy to satisfy a want, such as a book or a pen

Good is the working idea for this part of the chapter.

Service. an activity done for someone, such as binding a thesis or delivering parcels

Service is the working idea for this part of the chapter.

Scarcity. the condition in which resources are too few to satisfy all the wants people have

Scarcity is the working idea for this part of the chapter.

Choice. picking one option and giving up the others because resources cannot cover every option

Choice is the working idea for this part of the chapter.

Choice and Opportunity Cost is understood by separating the question into small steps.

Choice and Opportunity Cost is understood by separating the question into small steps.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Good** means a physical thing people buy to satisfy a want, such as a book or a pen. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in choice and opportunity cost, before moving to the next step.

The term **Service** means an activity done for someone, such as binding a thesis or delivering parcels. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in choice and opportunity cost, before moving to the next step.

The term **Scarcity** means the condition in which resources are too few to satisfy all the wants people have. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in choice and opportunity cost, before moving to the next step.

The term **Choice** means picking one option and giving up the others because resources cannot cover every option. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in choice and opportunity cost, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because

an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in choice and opportunity cost to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

1.3 | THE BASIC PROBLEMS OF AN ECONOMIC ORGANISATION | FRAMEWORK | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Opportunity cost** means the value of the best alternative given up when a choice is made. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the basic problems of an economic organisation, before moving to the next step.

The term **Firm** means an organisation that uses resources to make goods or services and sell them. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the basic problems of an economic organisation, before moving to the next step.

The term **Household** means a person or group living together that buys goods and services and supplies resources. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the basic problems of an economic organisation, before moving to the next step.

The term **Economic organisation** means the way a society or business arranges who decides what is made, how and for whom. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state

The section closes by restating the condition that controls the result.

This section asks what the basic problems of an economic organisation means before applying it.

Opportunity cost. the value of the best alternative given up when a choice is made

Opportunity cost is the working idea for this part of the chapter.

Firm. an organisation that uses resources to make goods or services and sell them

Firm is the working idea for this part of the chapter.

Household. a person or group living together that buys goods and services and supplies resources

Household is the working idea for this part of the chapter.

Economic organisation. the way a society or business arranges who decides what is made, how and for whom

Economic organisation is the working idea for this part of the chapter.

The Basic Problems of an Economic Organisation is understood by separating the question into small steps.

The Basic Problems of an Economic Organisation is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what defining economics means before applying it.

Economic system. the set of rules and institutions a country uses to answer its basic economic problems

Economic system is the working idea for this part of the chapter.

Defining Economics is understood by separating the question into small steps.

Defining Economics is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

precisely what is being measured or compared in the basic problems of an economic organisation, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in the basic problems of an economic organisation to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

1.4 | DEFINING ECONOMICS | DEFINING THEORY | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Economic system** means the set of rules and institutions a country uses to answer its basic economic problems. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in defining economics, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in defining economics is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in defining economics to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion

automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

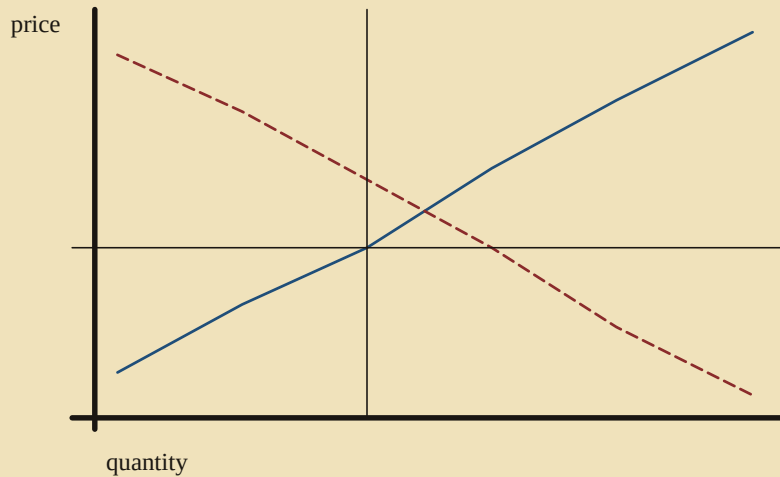


FIG. 1.1. a pair of bars comparing the forgone return of two stock choices, with the chosen option marked

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons is an illustrative documented case file. Base facts (FIXED): a bookshop in Dhaka opened on 4 January 2004 by Abdul Meghna; in 2026 run by his daughter Shirin Meghna and his son Imran Meghna; one shop of 600 square feet; two paid assistants. Record of 12 January 2026: restock fund Tk 400,000 in cash from family savings; shelf space allows one large restock only. Option A, textbooks: purchase cost Tk 400,000; expected sales Tk 560,000; expected sales less purchase cost Tk 160,000. Option B, stationery: purchase cost Tk 400,000; expected sales Tk 540,000; expected sales less purchase cost Tk 140,000. Decision recorded: option A. Opportunity cost of that decision: Tk 140,000. The three problems in shop terms: what to stock (textbooks), how to run the shop (family at the counter plus two assistants), for whom (university students first, schools second). Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact,

The worked case applies the chapter without adding facts to the record.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

The chapter has built one connected chain of reasoning.

Show why every society and every shop must choose, and give the reader the first working words of economics. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

The chapter uses a small number of historical references to anchor its defining ideas.

SUMMARY

THINK IT THROUGH

1.T1 scarcity versus shortage, forgone value of a choice made from a new price list, which of the three problems a given decision belongs to

1.T2 a student has Tk 9,000 and one free Saturday and must pick between two paid tasks and a course fee — name the opportunity cost

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to US petrol shortage after the October 1973 oil embargo, with queues and the odd-even plate rationing of 1974; search: 1973 oil embargo gasoline lines odd-even rationing. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

REVIEW QUESTIONS

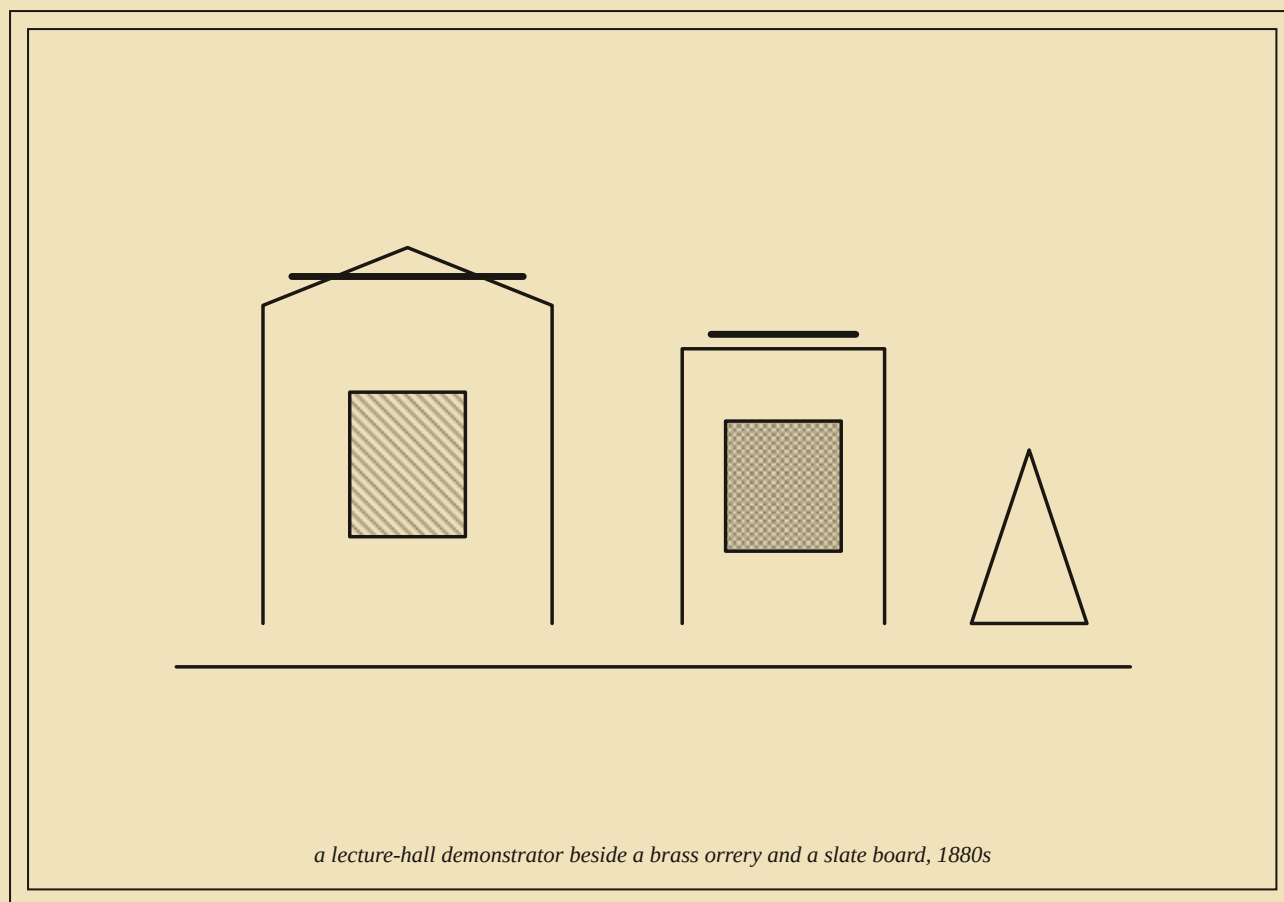
1.3 scarcity versus shortage, forgone value of a choice made from a new price list, which of the three problems a given decision belongs to

1.4 a student has Tk 9,000 and one free Saturday and must pick between two paid tasks and a course fee — name the opportunity cost

1.5 after the scarcity passage and after the opportunity-cost passage

PAUSE AND CHECK 1.P6 after the scarcity passage

PAUSE AND CHECK 1.P7 the opportunity-cost passage



CHAPTER II

Economic Reasoning, Laws and the Scope of Microeconomics

What does this chapter let us explain that the earlier ideas could not explain alone?



Teach the reader how economists argue, what an economic law is, and where microeconomics fits. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Teach the reader how economists argue, what an economic law is, and where microeconomics fits. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at Tk 60 and 200 copies sell in a week. Ledger note A, in Shirin Meghna's hand: when the price was raised to Tk 66 on 2 March 2026, weekly sales fell to 188. Ledger note B, in Imran Meghna's hand: students should not have to pay more than Tk 60. Note A can be checked against sales records; note B cannot. Regular tendency recorded over three years: textbook sales rise in the two weeks before each term begins (this holds when the shop's prices stay unchanged). Micro and macro: the shop's price and stock are micro matters; the movement of the average price of all goods in Bangladesh is a macro matter. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

2.1 | SCIENCE, ART AND NORMATIVE VIEWS | NEW IDEA | 550

This section asks what science, art and normative views means before applying it.

Science. a body of knowledge built by observing facts, forming explanations and testing them

Science is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Science** means a body of knowledge built by observing facts, forming explanations and testing them. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in science, art and normative views, before moving to the next step.

The term **Art** means the skilled use of knowledge to reach a practical aim. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in science, art and normative views, before moving to the next step.

The term **Positive statement** means a claim about what is, which can be checked against facts. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in science, art and normative views, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in science, art and normative views to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

2.2 | ECONOMIC LAWS | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Normative statement** means a claim about what ought to be, resting on values and opinion. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in economic laws, before moving to the next step.

Art. the skilled use of knowledge to reach a practical aim

Art is the working idea for this part of the chapter.

Positive statement. a claim about what is, which can be checked against facts

Positive statement is the working idea for this part of the chapter.

Science, Art and Normative Views is understood by separating the question into small steps.

Science, Art and Normative Views is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what economic laws means before applying it.

Normative statement. a claim about what ought to be, resting on values and opinion

Normative statement is the working idea for this part of the chapter.

Ceteris paribus. a Latin phrase meaning other things being equal, used to study one change at a time

Ceteris paribus is the working idea for this part of the chapter.

Economic law. a statement of a regular tendency in human choices, true when other things stay equal

Economic law is the working idea for this part of the chapter.

Economic Laws is understood by separating the question into small steps.

Economic Laws is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The term **Ceteris paribus** means a Latin phrase meaning other things being equal, used to study one change at a time. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in economic laws, before moving to the next step.

The term **Economic law** means a statement of a regular tendency in human choices, true when other things stay equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in economic laws, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in economic laws to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

2.3 | MICROECONOMICS AND MACROECONOMICS | FRAMEWORK | 450

This section asks what microeconomics and macroeconomics means before applying it.

Model. a simplified picture of a situation that keeps only the features needed to answer a question

Model is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Model** means a simplified picture of a situation that keeps only the features needed to answer a question. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or

compared in microeconomics and macroeconomics, before moving to the next step.

The term **Microeconomics** means the study of choices made by single households, firms and markets. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in microeconomics and macroeconomics, before moving to the next step.

The term **Macroeconomics** means the study of the economy as a whole, such as total output, jobs and the general price level. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in microeconomics and macroeconomics, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in microeconomics and macroeconomics is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in microeconomics and macroeconomics to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

2.4 | SCOPE, IMPORTANCE AND OBJECTIVES | FRAMEWORK | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

Microeconomics. the study of choices made by single households, firms and markets

Microeconomics is the working idea for this part of the chapter.

Macroeconomics. the study of the economy as a whole, such as total output, jobs and the general price level

Macroeconomics is the working idea for this part of the chapter.

Microeconomics and Macroeconomics is understood by separating the question into small steps.

Microeconomics and Macroeconomics is understood by separating the question into small steps.

Microeconomics and Macroeconomics is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what scope, importance and objectives means before applying it.

Scope, Importance and Objectives is understood by separating the question into small steps.

Scope, Importance and Objectives is understood by separating the question into small steps.

Scope, Importance and Objectives is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in scope, importance and objectives is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in scope, importance and objectives to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

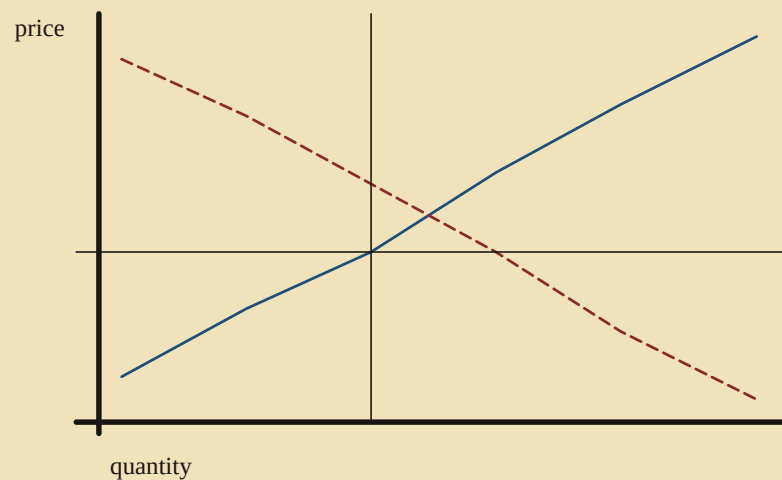


FIG. 2.1. a two-branch map of economics showing the questions that belong to microeconomics and to macroeconomics

The same method remains useful when the surrounding details change.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The same method remains useful when the surrounding details change.

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The same method remains useful when the surrounding details change.

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Base facts carried: bookshop in Dhaka, run in 2026 by Shirin and Imran Meghna, restock decision of 12 January 2026 (Chapter 1 facts need not be repeated beyond this). Records of 5 February 2026: a 200-page exercise book sells at Tk 60 and 200 copies sell in a week. Ledger note A, in Shirin Meghna's hand: when the price was raised to Tk 66 on 2 March 2026, weekly sales fell to 188. Ledger note B, in Imran Meghna's hand: students should not have to pay more than Tk 60. Note A can be checked against sales records; note B cannot. Regular tendency recorded over three years: textbook sales rise in the two weeks before each term begins (this holds when the shop's prices stay unchanged). Micro and macro: the shop's price and stock are micro matters; the movement of the average price of all goods in Bangladesh is a macro matter. **Question.** What does the record show when the chapter's central rule is applied to these facts?

The worked case applies the chapter without adding facts to the record.

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

The reasoning follows the data rather than an invented story.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

The conclusion is limited to what the case supports.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

A final check asks whether the result fits the original data.

* * *

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

SUMMARY

Teach the reader how economists argue, what an economic law is, and where microeconomics fits. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

2.T1 sort five statements into positive and normative, state a law with its other-things-equal condition, place four decisions under micro or macro

2.T2 a minister says bread should be cheaper — separate the checkable part from the opinion

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to Great Debasement under Henry VIII, 1544 to 1551, and Thomas Gresham's observation about money; search: Great Debasement Gresham's law. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

REVIEW QUESTIONS

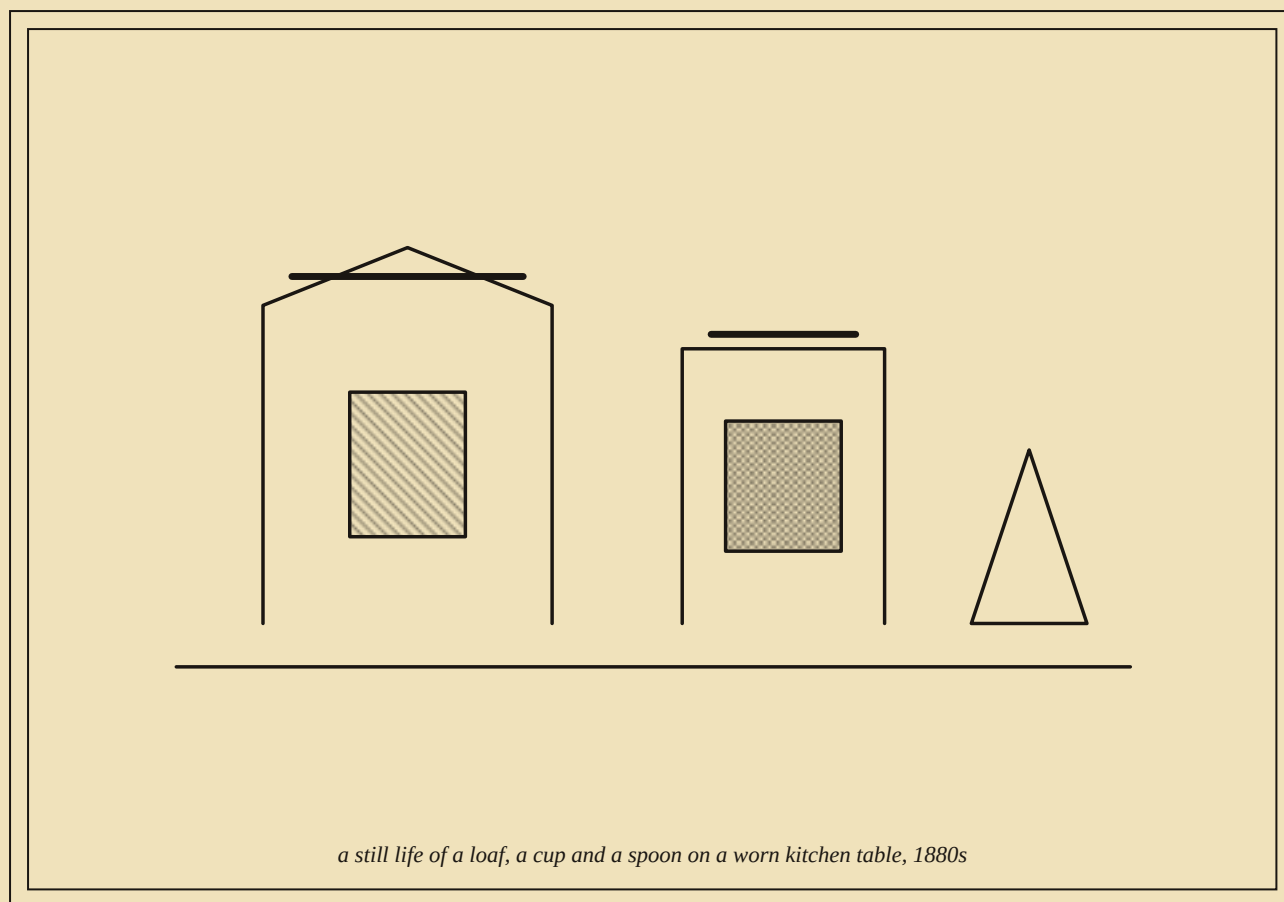
2.3 sort five statements into positive and normative, state a law with its other-things-equal condition, place four decisions under micro or macro

2.4 a minister says bread should be cheaper — separate the checkable part from the opinion

2.5 after the other-things-equal passage and after the micro-macro split

PAUSE AND CHECK 2.P6 after the other-things-equal passage

PAUSE AND CHECK 2.P7 the micro-macro split



CHAPTER III

Utility and Diminishing Marginal Utility

What does this chapter let us explain that the earlier ideas could not explain alone?



Explain satisfaction as the root of demand and give the first law the reader can test. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Explain satisfaction as the root of demand and give the first law the reader can test. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): total utility 20, 34, 43, 48, 50; marginal utility 20, 14, 9, 5, 2. A sixth notebook is rated 0 and a seventh would lower satisfaction by 3. Conditions noted on the card: all notebooks of the same kind, bought in a single visit, no change in taste. Notebook price Tk 60 (carried from the shop's price list). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

3.1 | UTILITY AND SATISFACTION | NEW IDEA | 775

This section asks what utility and satisfaction means before applying it.

Utility. the satisfaction a person gets from using a good or service

Utility is the working idea for this part of the chapter.

Consumer. a person who buys or uses goods and services to meet wants

Consumer is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Utility** means the satisfaction a person gets from using a good or service. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in utility and satisfaction, before moving to the next step.

The term **Consumer** means a person who buys or uses goods and services to meet wants. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is

being measured or compared in utility and satisfaction, before moving to the next step.

The term **Total utility** means the whole satisfaction from all the units of a good consumed in a period. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in utility and satisfaction, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in utility and satisfaction to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

3.2 | THE LAW OF DIMINISHING MARGINAL UTILITY | DEFINING THEORY | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Marginal utility** means the extra satisfaction gained from consuming one more unit of a good. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of diminishing marginal utility, before moving to the next step.

Total utility, the whole satisfaction from all the units of a good consumed in a period

Total utility is the working idea for this part of the chapter.

Utility and Satisfaction is understood by separating the question into small steps.

Utility and Satisfaction is understood by separating the question into small steps.

Utility and Satisfaction is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what the law of diminishing marginal utility means before applying it.

Marginal utility, the extra satisfaction gained from consuming one more unit of a good

Marginal utility is the working idea for this part of the chapter.

Util. an imaginary unit used to count satisfaction in examples

Util is the working idea for this part of the chapter.

Cardinal measurement.

measuring satisfaction in numbers, such as 20 utils, so amounts can be added and compared

Cardinal measurement is the working idea for this part of the chapter.

The Law of Diminishing Marginal Utility is understood by separating the question into small steps.

The Law of Diminishing Marginal Utility is understood by separating the question into small steps.

The Law of Diminishing Marginal Utility is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what limitations of the law means before applying it.

The term **Util** means an imaginary unit used to count satisfaction in examples. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of diminishing marginal utility, before moving to the next step.

The term **Cardinal measurement** means measuring satisfaction in numbers, such as 20 utils, so amounts can be added and compared. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of diminishing marginal utility, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in the law of diminishing marginal utility to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

3.3 | LIMITATIONS OF THE LAW | FRAMEWORK | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Ordinal measurement** means ranking satisfaction as first, second and third without giving it a number. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in limitations of the law, before moving to the next step.

The term **Law of diminishing marginal utility** means as a person consumes more units of a good, the extra satisfaction from each further unit falls. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in limitations of the law, before moving to the next step.

The term **Saturation point** means the quantity at which marginal utility reaches zero and total utility stops rising. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): t. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in limitations of the law, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in limitations of the law is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in limitations of the law to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

Ordinal measurement. ranking satisfaction as first, second and third without giving it a number

Ordinal measurement is the working idea for this part of the chapter.

Law of diminishing marginal utility. as a person consumes more units of a good, the extra satisfaction from each further unit falls

Law of diminishing marginal utility is the working idea for this part of the chapter.

Saturation point. the quantity at which marginal utility reaches zero and total utility stops rising

Saturation point is the working idea for this part of the chapter.

Limitations of the Law is understood by separating the question into small steps.

Limitations of the Law is understood by separating the question into small steps.

Limitations of the Law is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

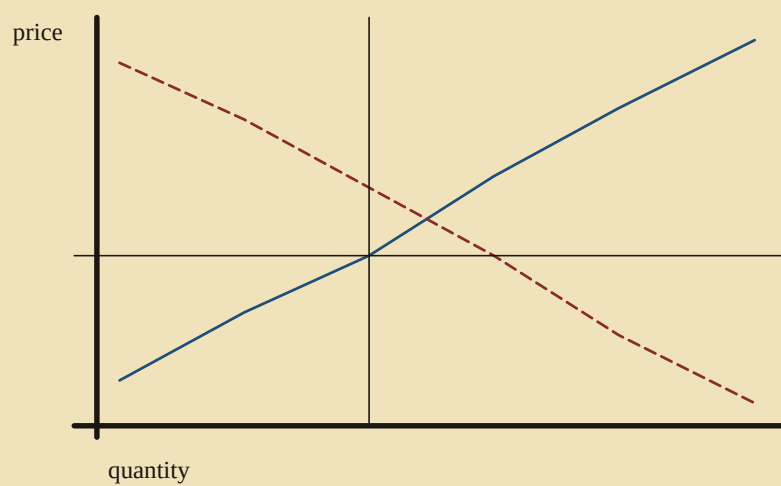


FIG. 3.1. *the total utility curve rising and flattening against units consumed*

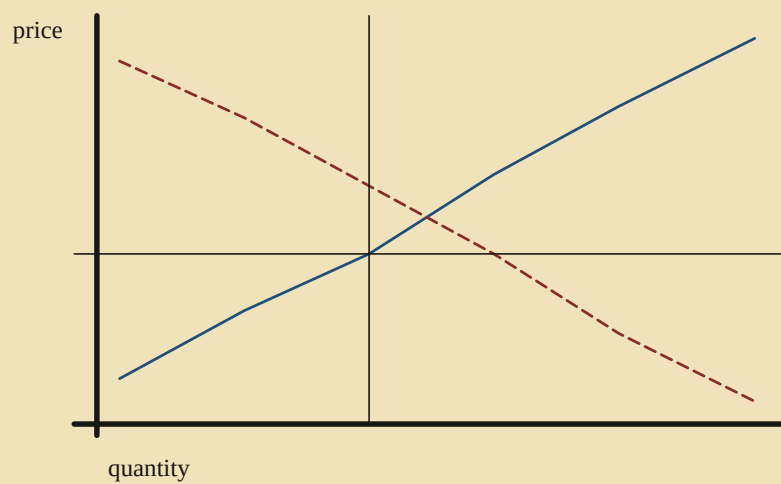


FIG. 3.2. *the marginal utility curve falling to zero against the same units*

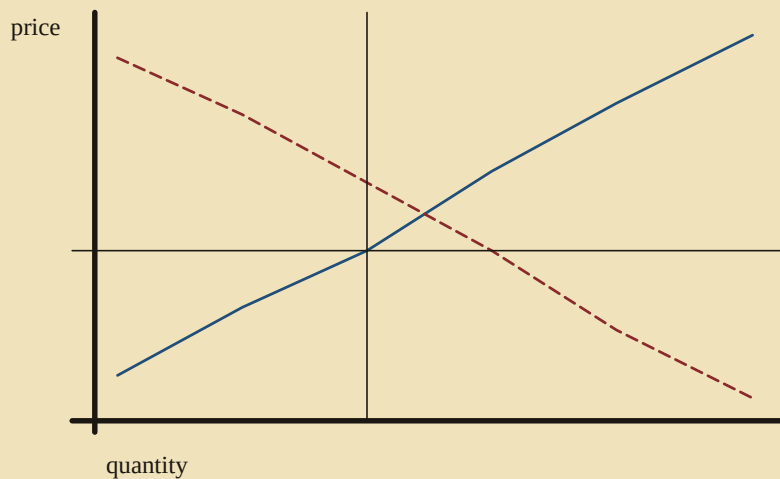


FIG. 3.3. one panel pairing total and marginal utility so the reader sees their link

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

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The same method remains useful when the surrounding details change.

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The same method remains useful when the surrounding details change.

The same method remains useful when the surrounding details change.

The same method remains useful when the surrounding details change.

The worked case applies the chapter without adding facts to the record.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Survey card of 10 February 2026 for one regular customer, satisfaction scored in utils for notebooks bought in one visit: first notebook 20 utils, second 14, third 9, fourth 5, fifth 2. Derived (by code in the writer's check): total utility 20, 34, 43, 48, 50; marginal utility 20, 14, 9, 5, 2. A sixth notebook is rated 0 and a seventh would lower satisfaction by 3. Conditions noted on the card: all notebooks of the same kind, bought in a single visit, no change in taste. Notebook price Tk 60 (carried from the shop's price list). Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Explain satisfaction as the root of demand and give the first law the reader can test. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

3.11 complete a total-and-marginal table from new scores, state the law with its conditions, give one apparent exception and say why it is not one

3.T2 a diner eats from a buffet — find where marginal utility reaches zero and say what a sensible diner does next

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

IN HISTORY

The entry directs attention to The marginal revolution of 1871 to 1874: Jevons, Menger and Walras explain the diamond-and-water puzzle; search: marginal revolution 1871 Jevons Menger diamond water paradox. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The historical case shows the chapter idea in a documented setting.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

The chapter uses a small number of historical references to anchor its defining ideas.

REVIEW QUESTIONS

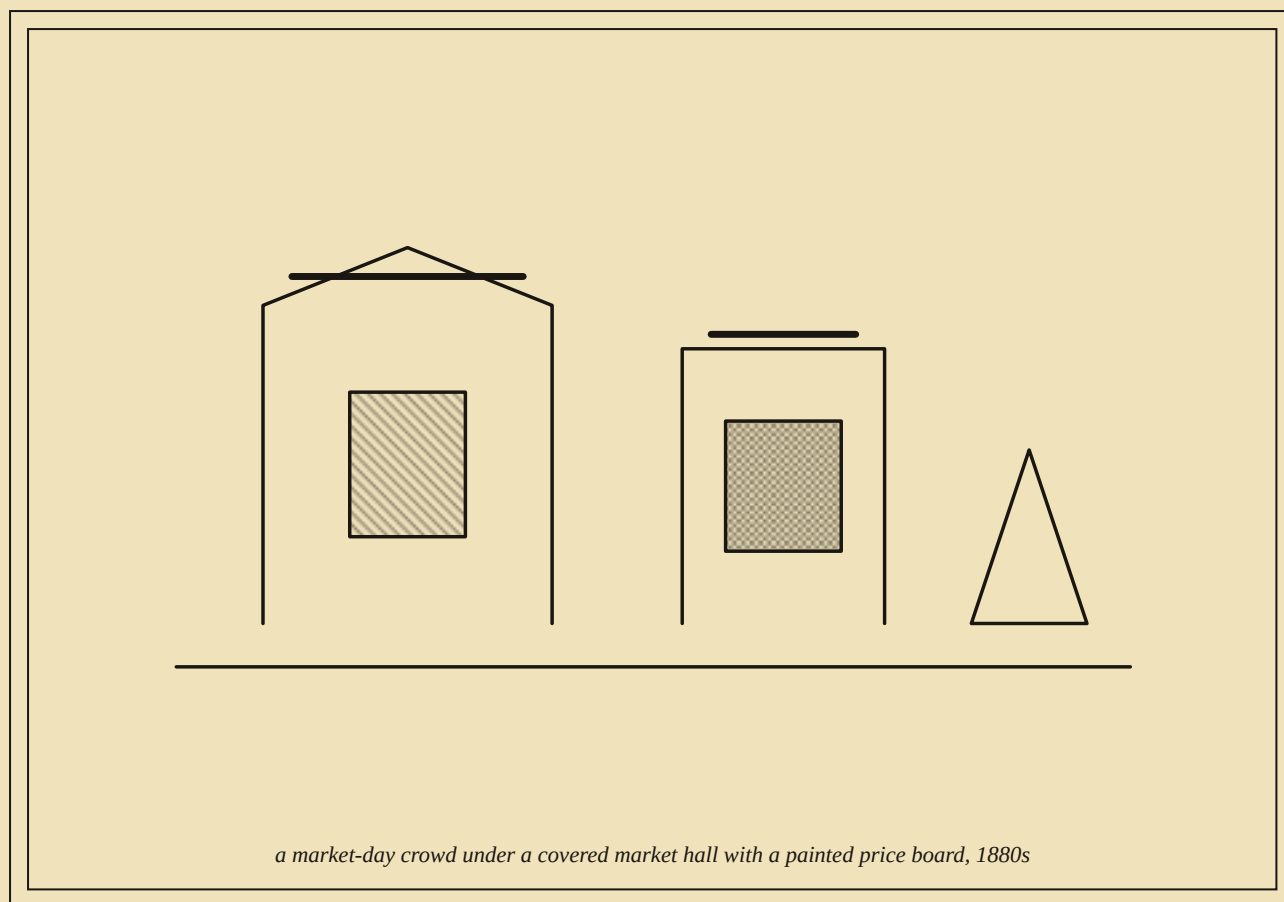
3.3 complete a total-and-marginal table from new scores, state the law with its conditions, give one apparent exception and say why it is not one

3.4 a diner eats from a buffet — find where marginal utility reaches zero and say what a sensible diner does next

3.5 after the total-versus-marginal passage and after the reasons-it-falls passage

PAUSE AND CHECK 3.P6 after the total-versus-marginal passage

PAUSE AND CHECK 3.P7 the reasons-it-falls passage



CHAPTER IV

Demand and the Demand Curve

What does this chapter let us explain that the earlier ideas could not explain alone?



Build the demand idea from a price table to a curve, and show what moves along it and what shifts it. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Build the demand idea from a price table to a curve, and show what moves along it and what shifts it. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188; Tk 70, 180. Shock to determinants: schools announced a drawing term on 10 March 2026, and weekly sales at Tk 60 rose from 200 to 230 (a shift of the curve). Complements: when pen-pack price was cut from Tk 100 to Tk 90 on 16 March 2026, notebook sales at Tk 60 rose from 200 to 210 a week. Substitutes: when the price of a second-hand prescribed textbook fell by 20 per cent, weekly sales of the new copy fell from 50 to 44 (new copy price Tk 700). Snob effect: a numbered edition at Tk 2,400 sold out in three days, while the unnumbered Tk 1,200 edition took three weeks. Bandwagon effect: a title at Tk 450 sold 20 copies a week until it topped a public bestseller list, then 85 a week. Giffen good: no case in the shop; explain only in general terms. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

4.1 | WHAT DEMAND MEANS | NEW IDEA | 450

This section asks what what demand means means before applying it.

Demand. the quantity of a good people are willing and able to buy at each possible price, in a given period

Demand is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Demand** means the quantity of a good people are willing and able to buy at each possible price, in a given period. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what demand means, before moving to the next step.

The term **Quantity demanded** means the amount people plan to buy at one particular price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what demand means, before moving to the next step.

The term **Demand schedule** means a table listing the quantity demanded at each of several prices. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what demand means, before moving to the next step.

The term **Demand curve** means a line on a graph showing the quantity demanded at each price, other things equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what demand means, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in what demand means to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

Quantity demanded. the amount people plan to buy at one particular price

Quantity demanded is the working idea for this part of the chapter.

Demand schedule. a table listing the quantity demanded at each of several prices

Demand schedule is the working idea for this part of the chapter.

Demand curve. a line on a graph showing the quantity demanded at each price, other things equal

Demand curve is the working idea for this part of the chapter.

What Demand Means is understood by separating the question into small steps.

What Demand Means is understood by separating the question into small steps.

What Demand Means is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

4.2 | THE LAW OF DEMAND | DEFINING THEORY |

550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of

This section asks what the law of demand means before applying it.

Market. any arrangement that brings buyers and sellers of a good together

Market is the working idea for this part of the chapter.

Market demand. the sum of all individual buyers' demands for a good at each price

Market demand is the working idea for this part of the chapter.

Law of demand. when the price of a good rises, quantity demanded falls, and when it falls, quantity demanded rises, other things equal

Law of demand is the working idea for this part of the chapter.

Determinants of demand. the things other than the good's own price that change how much people want to buy

Determinants of demand is the working idea for this part of the chapter.

The Law of Demand is understood by separating the question into small steps.

The Law of Demand is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Market** means any arrangement that brings buyers and sellers of a good together. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of demand, before moving to the next step.

The term **Market demand** means the sum of all individual buyers' demands for a good at each price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of demand, before moving to the next step.

The term **Law of demand** means when the price of a good rises, quantity demanded falls, and when it falls, quantity demanded rises, other things equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of demand, before moving to the next step.

The term **Determinants of demand** means the things other than the good's own price that change how much people want to buy. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of demand, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in the law of demand to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

4.3 | WHAT MOVES THE DEMAND CURVE | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Normal good** means a good for which demand rises when buyers' incomes rise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the demand curve, before moving to the next step.

The term **Inferior good** means a good for which demand falls when buyers' incomes rise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the demand curve, before moving to the next step.

The term **Substitute goods** means goods that can replace each other, so a rise in one's price raises demand for the other. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the demand curve, before moving to the next step.

The term **Complementary goods** means goods used together, so a rise in one's price lowers demand for the other. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the demand curve, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states

This section asks what what moves the demand curve means before applying it.

Normal good. a good for which demand rises when buyers' incomes rise

Normal good is the working idea for this part of the chapter.

Inferior good. a good for which demand falls when buyers' incomes rise

Inferior good is the working idea for this part of the chapter.

Substitute goods. goods that can replace each other, so a rise in one's price raises demand for the other

Substitute goods is the working idea for this part of the chapter.

Complementary goods. goods used together, so a rise in one's price lowers demand for the other

Complementary goods is the working idea for this part of the chapter.

What Moves the Demand Curve is understood by separating the question into small steps.

What Moves the Demand Curve is understood by separating the question into small steps.

the

The section closes by restating the condition that controls the result.

condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in what moves the demand curve to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

This section asks what related goods and exceptions means before applying it.

Giffen good. a rare inferior good for which a price rise leads buyers to buy more of it

Giffen good is the working idea for this part of the chapter.

Snob effect. a fall in demand for a good because more people begin to own it

Snob effect is the working idea for this part of the chapter.

Bandwagon effect. a rise in demand for a good because more people are seen to buy it

Bandwagon effect is the working idea for this part of the chapter.

Related Goods and Exceptions is understood by separating the question into small steps.

Related Goods and Exceptions is understood by separating the question into small steps.

4.4 | RELATED GOODS AND EXCEPTIONS | NEW IDEA | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Giffen good** means a rare inferior good for which a price rise leads buyers to buy more of it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in related goods and exceptions, before moving to the next step.

The term **Snob effect** means a fall in demand for a good because more people begin to own it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in related goods and exceptions, before moving to the next step.

The term **Bandwagon effect** means a rise in demand for a good because more people are seen to buy it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in related goods and exceptions, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in related goods and exceptions is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The

result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in related goods and exceptions to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

Related Goods and Exceptions is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

* * *

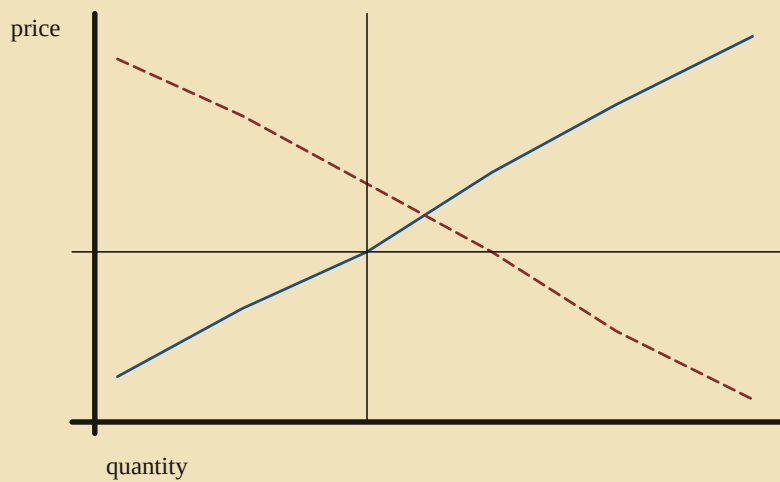


FIG. 4.1. a demand curve drawn from the exercise-book schedule

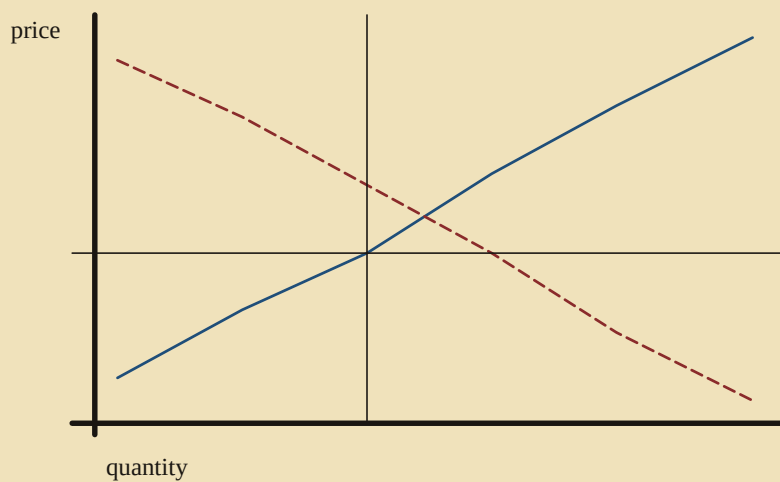


FIG. 4.2. a movement along the curve set beside a shift of the whole curve

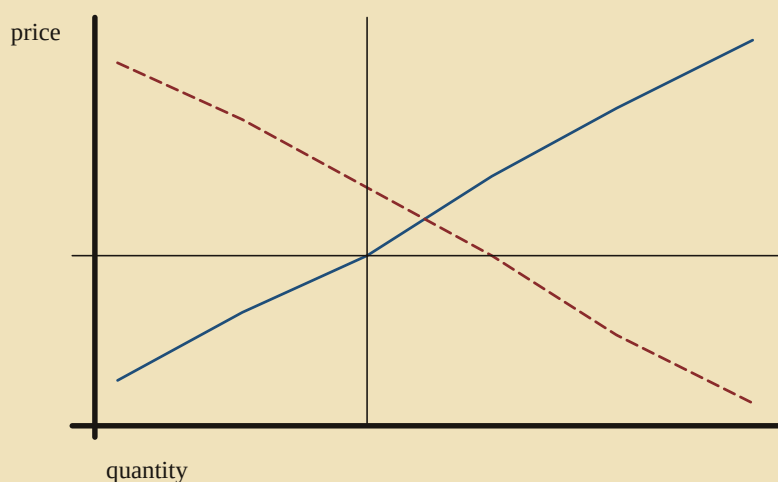


FIG. 4.3. a rightward and a leftward shift with the causes labelled

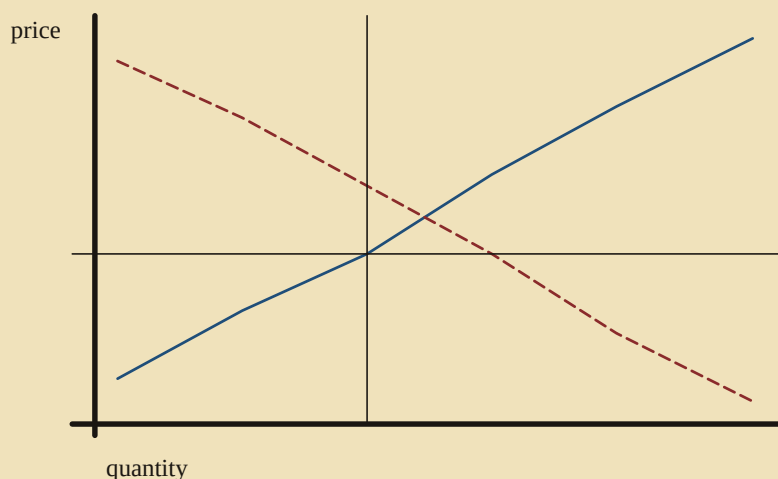


FIG. 4.4. the curve of an exceptional good that bends upward, with the snob and bandwagon cases beside it

WORKED CASE: THE RUNNING RECORD

The worked case applies the chapter without adding facts to the record.

Situation. Meghna and Sons (illustrative). Five-week trial from 26 January to 27 February 2026, one price per week, 200-page exercise book. Weekly sales: Tk 50, 240 copies; Tk 55, 220; Tk 60, 200; Tk 66, 188; Tk 70, 180. Shock to determinants: schools announced a drawing term on 10 March 2026, and weekly sales at Tk 60 rose from 200 to 230 (a shift of the curve). Complements: when pen-pack price was cut from Tk 100 to Tk 90 on 16 March 2026, notebook sales at Tk 60 rose from 200 to 210 a week. Substitutes: when the price of a second-hand prescribed textbook fell by 20 per cent, weekly sales of the new copy fell from 50 to 44 (new copy price Tk 700). Snob effect: a numbered edition at Tk 2,400 sold out in three days, while the unnumbered Tk 1,200 edition took three weeks. Bandwagon effect: a title at Tk 450 sold 20 copies a week until it topped a public bestseller list, then 85 a week. Giffen good: no case in the shop; explain only in general terms. Question.

What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

The reasoning follows the data rather than an invented story.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

The conclusion is limited to what the case supports.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

A final check asks whether the result fits the original data.

* * *

SUMMARY

Build the demand idea from a price table to a curve, and show what moves along it and what shifts it. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

The chapter has built one connected chain of reasoning.

THINK IT THROUGH

4.T1 draw a curve from a new schedule, tell a movement from a shift in five events, say how a pen-pack price change affects notebook demand

4.T2 when the price of petrol rises, say what probably happens to demand for natural gas and why

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

A common error is to apply a rule after its condition has changed.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

Another error is to confuse a numerical result with the meaning of that result.

IN HISTORY

The entry directs attention to Gregory King's table of harvest shortfalls and wheat prices, published by Charles Davenant in 1699; search: Gregory King Davenant 1699 harvest wheat price. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The historical case shows the chapter idea in a documented setting.

REVIEW QUESTIONS

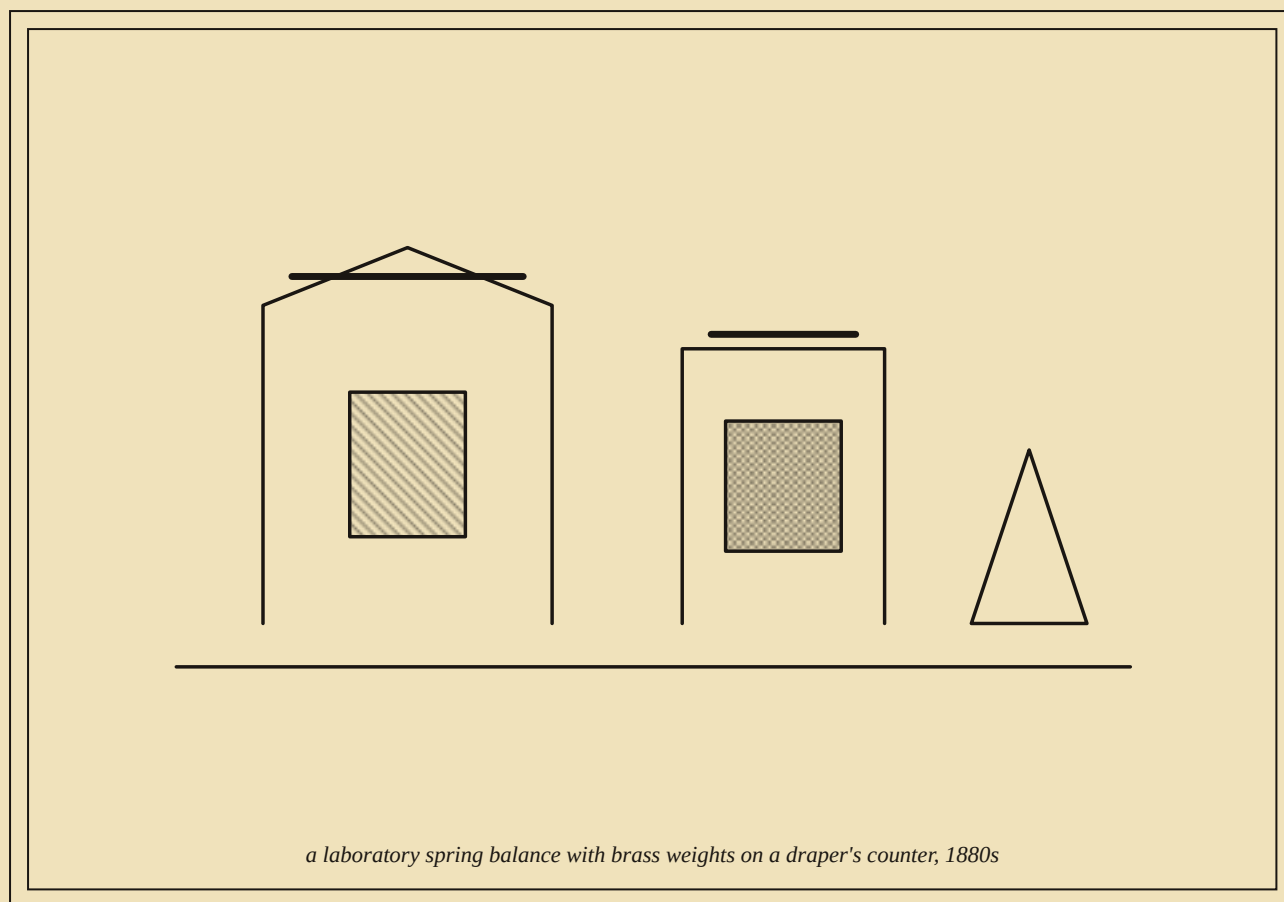
4.3 draw a curve from a new schedule, tell a movement from a shift in five events, say how a pen-pack price change affects notebook demand

4.4 when the price of petrol rises, say what probably happens to demand for natural gas and why

4.5 after the movement-versus-shift passage and after the exceptions passage

PAUSE AND CHECK 4.P6 after the movement-versus-shift passage

PAUSE AND CHECK 4.P7 the exceptions passage



CHAPTER V

Elasticity of Demand

What does this chapter let us explain that the earlier ideas could not explain alone?



Teach how to measure how strongly buyers respond, with percentages, and what the answer means for revenue. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Teach how to measure how strongly buyers respond, with percentages, and what the answer means for revenue. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic); total revenue Tk 12,000 (60 x 200) to Tk 12,408 (66 x 188). Messier rung: midpoint method gives about 0.65. Second good: a Tk 300 atlas, price cut to Tk 270 raised weekly sales from 20 to 30: -10 per cent price, +50 per cent quantity, elasticity 5 in size (elastic); revenue Tk 6,000 to Tk 8,100. Income elasticity: a household income rose from Tk 30,000 to Tk 33,000 a month and its novels bought rose from 50 to 60 a year: +10 per cent and +20 per cent, elasticity 2. Cross elasticity: a rival shop raised its notebook price from Tk 50 to Tk 55 (+10 per cent) and Meghna and Sons sales of its own notebooks rose from 100 to 108 a week (+8 per cent), cross elasticity 0.8. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

5.1 | MEASURING RESPONSIVENESS | TECHNIQUE | 775

This section asks what measuring responsiveness means before applying it.

Elasticity, a measure of how strongly one quantity responds when another quantity changes

Elasticity is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Elasticity** means a measure of how strongly one quantity responds when another quantity changes. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in measuring responsiveness, before moving to the next step.

The term **Percentage change** means the size of a change divided by the starting value, multiplied by 100. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in measuring responsiveness, before moving to the next step.

The term **Price elasticity of demand** means the percentage change in quantity demanded divided by the percentage change in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in measuring responsiveness, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in measuring responsiveness to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

Percentage change. the size of a change divided by the starting value, multiplied by 100

Percentage change is the working idea for this part of the chapter.

Price elasticity of demand. the percentage change in quantity demanded divided by the percentage change in price

Price elasticity of demand is the working idea for this part of the chapter.

Measuring Responsiveness is understood by separating the question into small steps.

Measuring Responsiveness is understood by separating the question into small steps.

Measuring Responsiveness is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

5.2 | FORMS OF PRICE ELASTICITY | NEW IDEA | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

This section asks what forms of price elasticity means before applying it.

Elastic demand. demand for which the percentage fall in quantity is greater than the percentage rise in price

Elastic demand is the working idea for this part of the chapter.

Inelastic demand. demand for which the percentage fall in quantity is smaller than the percentage rise in price

Inelastic demand is the working idea for this part of the chapter.

Unit elastic demand. demand for which the percentage change in quantity equals the percentage change in price

Unit elastic demand is the working idea for this part of the chapter.

Forms of Price Elasticity is understood by separating the question into small steps.

Forms of Price Elasticity is understood by separating the question into small steps.

Forms of Price Elasticity is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The term **Elastic demand** means demand for which the percentage fall in quantity is greater than the percentage rise in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forms of price elasticity, before moving to the next step.

The term **Inelastic demand** means demand for which the percentage fall in quantity is smaller than the percentage rise in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forms of price elasticity, before moving to the next step.

The term **Unit elastic demand** means demand for which the percentage change in quantity equals the percentage change in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forms of price elasticity, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in forms of price elasticity to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

5.3 | WHAT AFFECTS PRICE ELASTICITY | NEW IDEA | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Perfectly elastic demand** means demand that vanishes completely for any price rise, however small, shown as a flat curve. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what affects price elasticity, before moving to the next step.

The term **Perfectly inelastic demand** means demand that does not change at all when price changes, shown as a vertical curve. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what affects price elasticity, before moving to the next step.

The term **Total revenue** means the money a seller receives from sales, found by price multiplied by quantity sold. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what affects price elasticity, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in what affects price elasticity to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

This section asks what what affects price elasticity means before applying it.

Perfectly elastic demand. demand that vanishes completely for any price rise, however small, shown as a flat curve

Perfectly elastic demand is the working idea for this part of the chapter.

Perfectly inelastic demand. demand that does not change at all when price changes, shown as a vertical curve

Perfectly inelastic demand is the working idea for this part of the chapter.

Total revenue. the money a seller receives from sales, found by price multiplied by quantity sold

Total revenue is the working idea for this part of the chapter.

What Affects Price Elasticity is understood by separating the question into small steps.

What Affects Price Elasticity is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

5.4 | INCOME AND CROSS ELASTICITY | TECHNIQUE

| 550

This section asks what income and cross elasticity means before applying it.

Income elasticity of demand. the percentage change in quantity demanded divided by the percentage change in buyers' income

Income elasticity of demand is the working idea for this part of the chapter.

Cross elasticity of demand. the percentage change in quantity demanded of one good divided by the percentage change in another good's price

Cross elasticity of demand is the working idea for this part of the chapter.

Income and Cross Elasticity is understood by separating the question into small steps.

Income and Cross Elasticity is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Income elasticity of demand** means the percentage change in quantity demanded divided by the percentage change in buyers' income. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in income and cross elasticity, before moving to the next step.

The term **Cross elasticity of demand** means the percentage change in quantity demanded of one good divided by the percentage change in another good's price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic). In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in income and cross elasticity, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in income and cross elasticity is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in income and cross elasticity to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

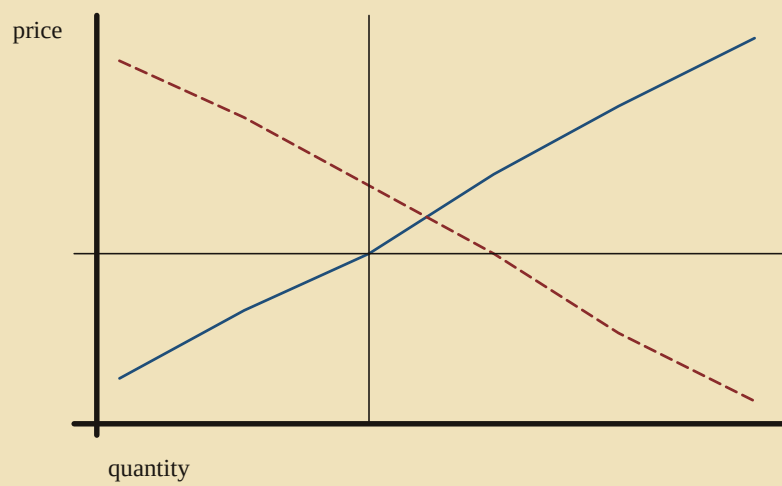


FIG. 5.1. *a steep and a flat demand curve side by side for the same price change*

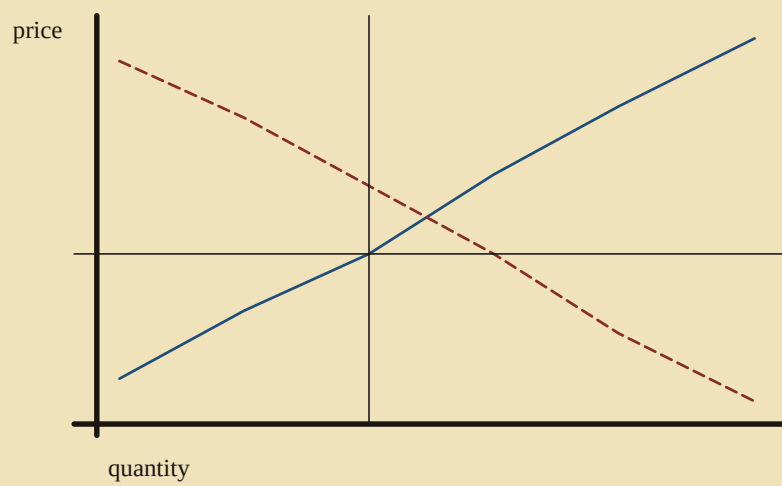


FIG. 5.2. *five demand curves for the five forms of price elasticity*

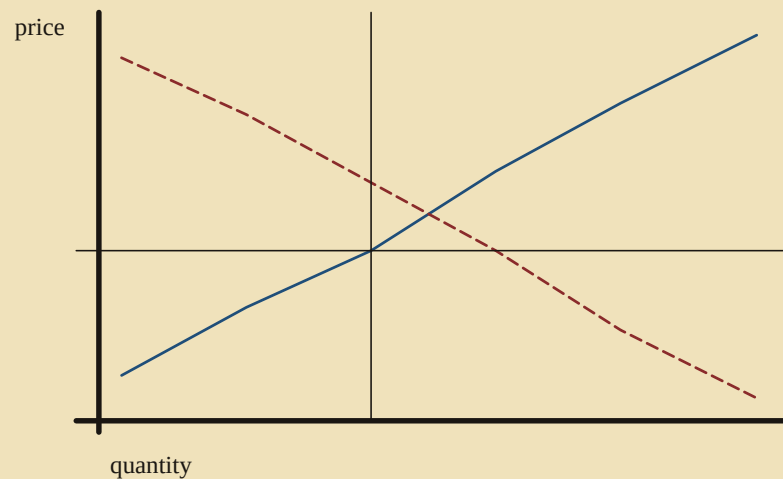


FIG. 5.3. total revenue before and after a price rise for an inelastic and an elastic good

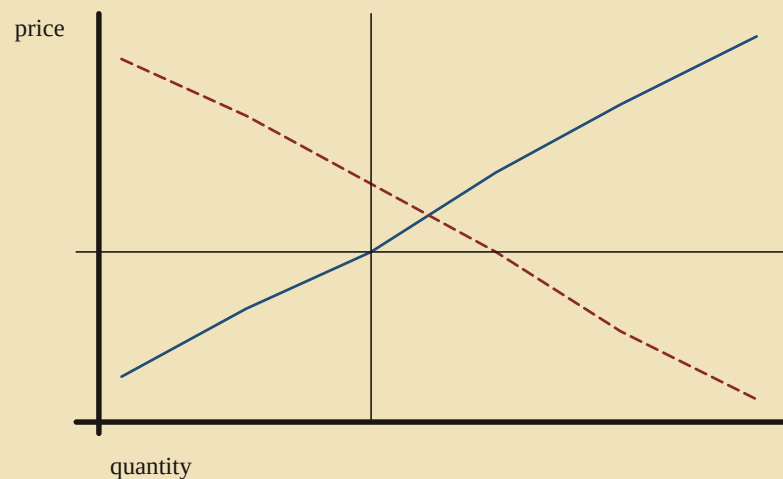


FIG. 5.4. a number line placing income elasticity values for necessities, comforts and luxuries

WORKED CASE: THE RUNNING RECORD

The worked case applies the chapter without adding facts to the record.

Situation. Meghna and Sons (illustrative). Price elasticity: 200-page exercise book, price rose from Tk 60 to Tk 66 on 2 March 2026; weekly sales fell from 200 to 188. Clean figures: price change +10 per cent, quantity change -6 per cent, elasticity 0.6 (inelastic); total revenue Tk 12,000 (60×200) to Tk 12,408 (66×188). Messier rung: midpoint method gives about 0.65. Second good: a Tk 300 atlas, price cut to Tk 270 raised weekly sales from 20 to 30: -10 per cent price, +50 per cent quantity, elasticity 5 in size (elastic); revenue Tk 6,000 to Tk 8,100. Income elasticity: a household income rose from Tk 30,000 to Tk 33,000 a month and its novels bought rose from 50 to 60 a year: +10 per cent and +20 per cent, elasticity 2. Cross elasticity: a rival shop raised its notebook price from Tk 50 to Tk 55 (+10 per cent) and Meghna and Sons sales of its own notebooks rose from 100 to 108 a week (+8 per cent), cross elasticity 0.8. Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

The reasoning follows the data rather than an invented story.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

The conclusion is limited to what the case supports.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

A final check asks whether the result fits the original data.

* * *

SUMMARY

Teach how to measure how strongly buyers respond, with percentages, and what the answer means for revenue. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

The chapter has built one connected chain of reasoning.

THINK IT THROUGH

5.T1 compute three elasticities from new price-quantity pairs and name each form, say which goods have low elasticity and why, compute income and cross elasticity from new data

5.T2 a seller wants more revenue — decide from the elasticity whether to cut or raise the price

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

A common error is to apply a rule after its condition has changed.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

Another error is to confuse a numerical result with the meaning of that result.

IN HISTORY

The entry directs attention to Alfred Marshall introduces elasticity of demand in Principles of Economics, 1890; search: Marshall Principles of Economics 1890 elasticity of demand origin. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The historical case shows the chapter idea in a documented setting.

The chapter uses a small number of historical references to anchor its defining ideas.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

REVIEW QUESTIONS

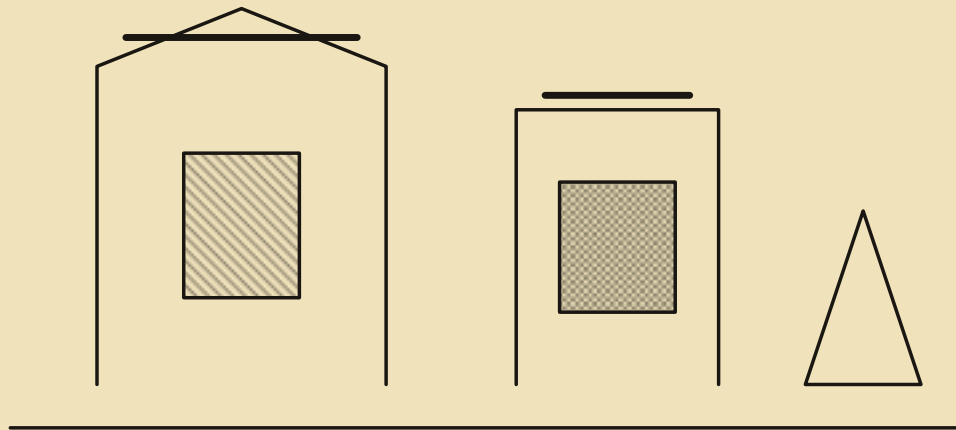
5.3 compute three elasticities from new price-quantity pairs and name each form, say which goods have low elasticity and why, compute income and cross elasticity from new data

5.4 a seller wants more revenue — decide from the elasticity whether to cut or raise the price

5.5 after the formula passage and after the revenue passage

PAUSE AND CHECK 5.P6 after the formula passage

PAUSE AND CHECK 5.P7 the revenue passage



a draper's shop with a customer weighing two bolts of cloth against a window, 1880s

CHAPTER VI

Consumer Choice: Surplus and Indifference Curves

What does this chapter let us explain that the earlier ideas could not explain alone?



Show what a buyer gains from a purchase and how a buyer with a fixed budget picks the best bundle. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Show what a buyer gains from a purchase and how a buyer with a fixed budget picks the best bundle. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650; four buy and one does not. Surpluses: Tk 250, Tk 150, Tk 50, Tk 0, total Tk 450. Choice: a student's monthly spending plan is Tk 1,200 on prescribed books at Tk 150 each and stationery packs at Tk 100 each. Budget line bundles (books, packs): (0, 12), (2, 9), (4, 6), (6, 3), (8, 0). One indifference curve through (4, 6) also passes through (3, 8) and (6, 4); the slope between (3, 8) and (4, 6) is 2 packs per book, between (4, 6) and (6, 4) it is 1 pack per book. Best bundle: (4, 6), cost Tk 600 plus Tk 600 equals Tk 1,200; the price ratio 150 to 100 equals 1.5 packs per book. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

6.1 | CONSUMPTION AND CONSUMER SURPLUS | NEW IDEA | 550

This section asks what consumption and consumer surplus means before applying it.

Consumption. the using up of goods and services to meet wants

Consumption is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Consumption** means the using up of goods and services to meet wants. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in consumption and consumer surplus, before moving to the next step.

The term **Willingness to pay** means the highest price a buyer would accept to get one unit of a good. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in consumption and consumer surplus, before moving to the next step.

The term **Consumer surplus** means the gap between what a buyer would pay and what is actually paid. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in consumption and consumer surplus, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in consumption and consumer surplus to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

6.2 | INDIFFERENCE CURVES AND THE BUDGET LINE | FRAMEWORK | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Indifference curve** means a line joining bundles of two goods that give a consumer the same satisfaction. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in indifference curves and the budget line, before moving to the next step.

Willingness to pay. the highest price a buyer would accept to get one unit of a good

Willingness to pay is the working idea for this part of the chapter.

Consumer surplus. the gap between what a buyer would pay and what is actually paid

Consumer surplus is the working idea for this part of the chapter.

Consumption and Consumer Surplus is understood by separating the question into small steps.

Consumption and Consumer Surplus is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what indifference curves and the budget line means before applying it.

Indifference curve. a line joining bundles of two goods that give a consumer the same satisfaction

Indifference curve is the working idea for this part of the chapter.

Indifference map. a set of indifference curves, with curves farther from the origin showing more satisfaction

Indifference map is the working idea for this part of the chapter.

Marginal rate of substitution. the amount of one good a consumer will give up to gain one more unit of the other

Marginal rate of substitution is the working idea for this part of the chapter.

Indifference Curves and the Budget Line is understood by separating the question into small steps.

Indifference Curves and the Budget Line is understood by separating the question into small steps.

Indifference Curves and the Budget Line is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what consumer equilibrium means before applying it.

Budget line. a line showing every bundle of two goods a consumer can exactly afford with a given income

Budget line is the working idea for this part of the chapter.

The term **Indifference map** means a set of indifference curves, with curves farther from the origin showing more satisfaction. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in indifference curves and the budget line, before moving to the next step.

The term **Marginal rate of substitution** means the amount of one good a consumer will give up to gain one more unit of the other. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in indifference curves and the budget line, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in indifference curves and the budget line to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

6.3 | CONSUMER EQUILIBRIUM | TECHNIQUE | 650

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Budget line** means a line showing every bundle of two goods a consumer can exactly afford with a given income. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk

650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in consumer equilibrium, before moving to the next step.

The term **Consumer equilibrium** means the affordable bundle that gives a consumer the greatest satisfaction. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in consumer equilibrium, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in consumer equilibrium is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in consumer equilibrium to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

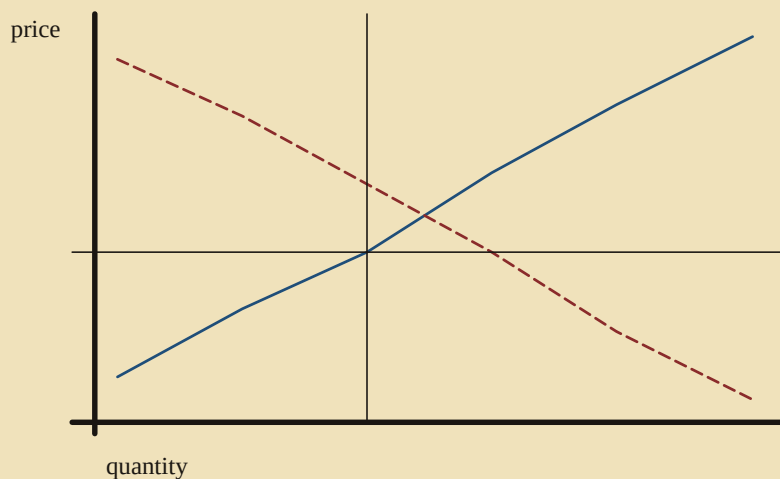


FIG. 6.1. a demand curve with the consumer-surplus triangle shaded

Consumer equilibrium. the affordable bundle that gives a consumer the greatest satisfaction

Consumer equilibrium is the working idea for this part of the chapter.

Consumer Equilibrium is understood by separating the question into small steps.

Consumer Equilibrium is understood by separating the question into small steps.

Consumer Equilibrium is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

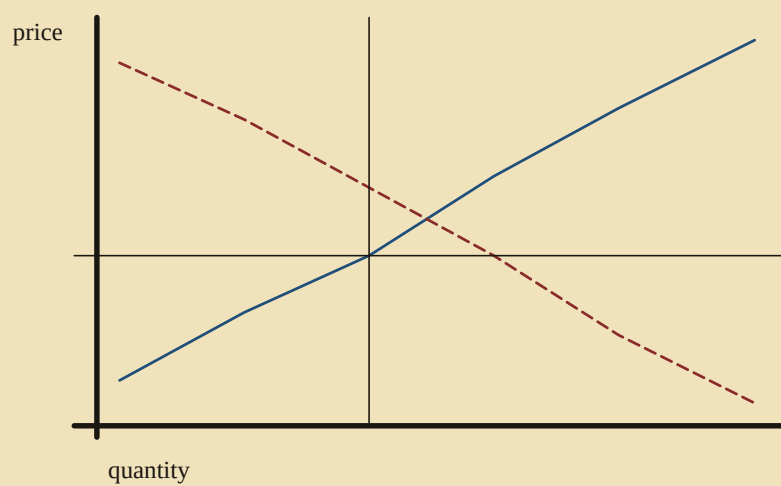


FIG. 6.2. *a family of indifference curves, convex to the origin*

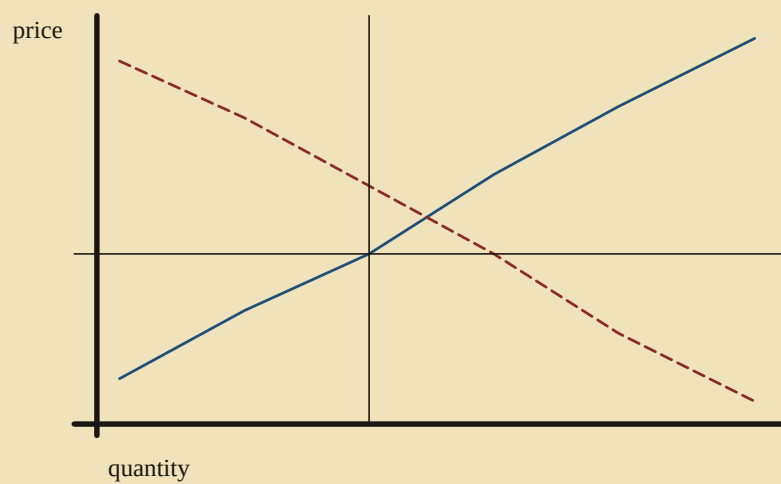


FIG. 6.3. *a budget line drawn from a spending plan of Tk 1,200*

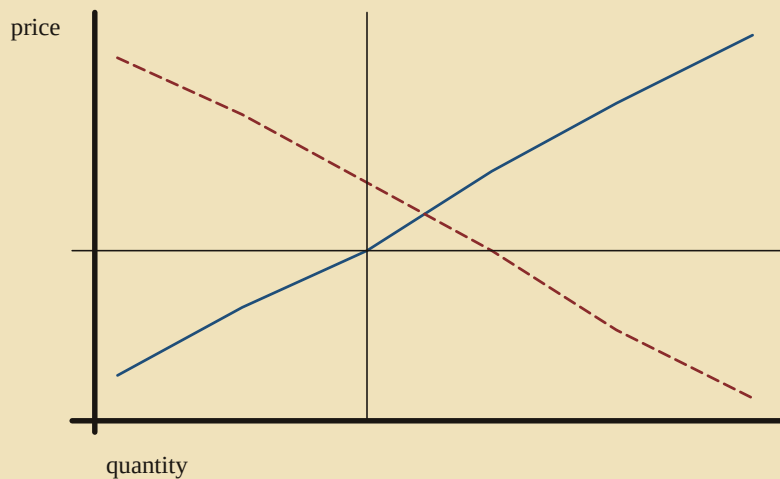


FIG. 6.4. the tangency point of curve and budget line marked as the best bundle

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

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The same method remains useful when the surrounding details change.

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The same method remains useful when the surrounding details change.

The same method remains useful when the surrounding details change.

The worked case applies the chapter without adding facts to the record.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

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WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Consumer surplus: on 14 March 2026 five customers' highest prices for a dictionary were Tk 900, Tk 800, Tk 700, Tk 650 and Tk 600; shelf price Tk 650; four buy and one does not. Surpluses: Tk 250, Tk 150, Tk 50, Tk 0, total Tk 450. Choice: a student's monthly spending plan is Tk 1,200 on prescribed books at Tk 150 each and stationery packs at Tk 100 each. Budget line bundles (books, packs): (0, 12), (2, 9), (4, 6), (6, 3), (8, 0). One indifference curve through (4, 6) also passes through (3, 8) and (6, 4); the slope between (3, 8) and (4, 6) is 2 packs per book, between (4, 6) and (6, 4) it is 1 pack per book. Best bundle: (4, 6), cost Tk 600 plus Tk 600 equals Tk 1,200; the price ratio 150 to 100 equals 1.5 packs per book. Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Show what a buyer gains from a purchase and how a buyer with a fixed budget picks the best bundle. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

The chapter has built one connected chain of reasoning.

THINK IT THROUGH

6.T1 compute consumer surplus from a new price list, draw a budget line from new prices, say why indifference curves do not cross

6.T2 a student's income doubles with prices unchanged — describe the new budget line and the likely best bundle

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

A common error is to apply a rule after its condition has changed.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

Another error is to confuse a numerical result with the meaning of that result.

IN HISTORY

The entry directs attention to Jules Dupuit, 1844, on measuring the usefulness of public works such as bridge tolls; search: Dupuit 1844 utility of public works bridge toll. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The historical case shows the chapter idea in a documented setting.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to accept a historical interpretation merely because it appears in a secondary account.

The chapter uses a small number of historical references to anchor its defining ideas.

REVIEW QUESTIONS

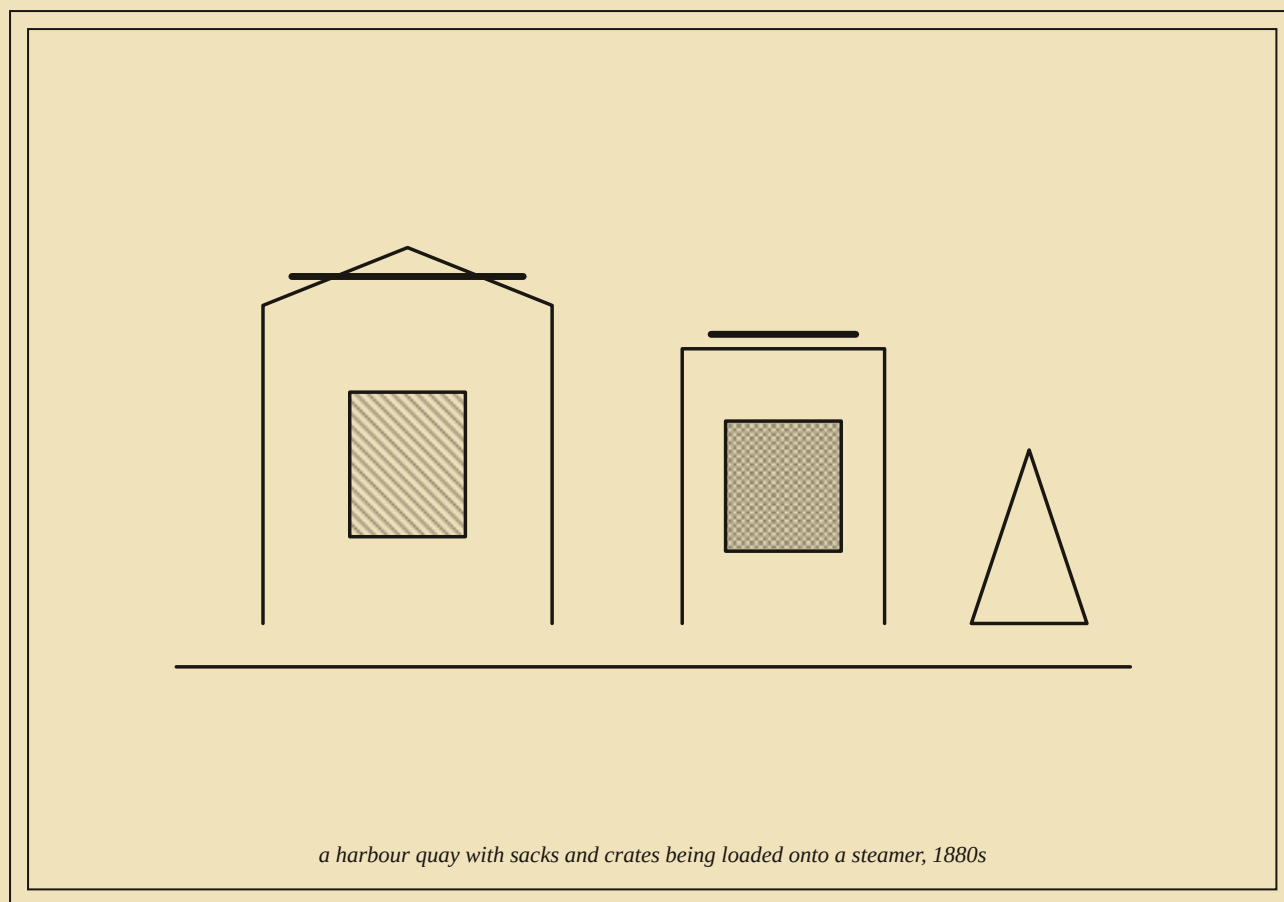
6.3 compute consumer surplus from a new price list, draw a budget line from new prices, say why indifference curves do not cross

6.4 a student's income doubles with prices unchanged — describe the new budget line and the likely best bundle

6.5 after the first indifference curve is drawn and after the tangency passage

PAUSE AND CHECK 6.P6 after the first indifference curve is drawn

PAUSE AND CHECK 6.P7 the tangency passage



a harbour quay with sacks and crates being loaded onto a steamer, 1880s

CHAPTER VII

Supply and Elasticity of Supply

What does this chapter let us explain that the earlier ideas could not explain alone?



Build the supply idea from a price table to a curve and measure how strongly sellers respond to price. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Build the supply idea from a price table to a curve and measure how strongly sellers respond to price. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300; Tk 60, 350. Elasticity from Tk 50 to Tk 55: +10 per cent price, +20 per cent quantity, elasticity 2.0 (elastic). Shift: on 1 April 2026 the printer's paper cost rose and it offered 50 fewer books at every price: 100, 150, 200, 250, 300. Exceptional curve 1: a rare first edition, only one copy exists, supply unchanged at any price. Exceptional curve 2: a part-time binder offers 20 hours a week at Tk 150 an hour, 26 hours at Tk 200, 24 hours at Tk 260. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

7.1 | THE LAW OF SUPPLY | DEFINING THEORY | 550

This section asks what the law of supply means before applying it.

Supply. the quantity of a good sellers are willing and able to offer at each possible price, in a given period

Supply is the working idea for this part of the chapter.

Quantity supplied. the amount sellers plan to offer at one particular price

Quantity supplied is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Supply** means the quantity of a good sellers are willing and able to offer at each possible price, in a given period. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of supply, before moving to the next step.

The term **Quantity supplied** means the amount sellers plan to offer at one particular price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader

state precisely what is being measured or compared in the law of supply, before moving to the next step.

The term **Supply schedule** means a table listing the quantity supplied at each of several prices. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the law of supply, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in the law of supply to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

7.2 | WHAT MOVES THE SUPPLY CURVE | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Supply curve** means a line on a graph showing the quantity supplied at each price, other things equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the supply curve, before moving to the next step.

The term **Law of supply** means when the price of a good rises, quantity supplied rises, and when it falls, quantity supplied falls, other things equal. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being

Supply schedule. a table listing the quantity supplied at each of several prices

Supply schedule is the working idea for this part of the chapter.

The Law of Supply is understood by separating the question into small steps.

The Law of Supply is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what what moves the supply curve means before applying it.

Supply curve. a line on a graph showing the quantity supplied at each price, other things equal

Supply curve is the working idea for this part of the chapter.

Law of supply. when the price of a good rises, quantity supplied rises, and when it falls, quantity supplied falls, other things equal

Law of supply is the working idea for this part of the chapter.

Determinants of supply. the things other than the good's own price that change how much sellers offer

Determinants of supply is the working idea for this part of the chapter.

What Moves the Supply Curve is understood by separating the question into small steps.

What Moves the Supply Curve is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

measured or compared in what moves the supply curve, before moving to the next step.

The term **Determinants of supply** means the things other than the good's own price that change how much sellers offer. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in what moves the supply curve, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in what moves the supply curve to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

7.3 | EXCEPTIONS TO THE LAW | FRAMEWORK | 225

This section asks what exceptions to the law means before applying it.

Backward-bending supply curve. a curve showing that beyond some pay level a worker offers fewer hours as pay rises further

Backward-bending supply curve is the working idea for this part of the chapter.

Price elasticity of supply. the percentage change in quantity supplied divided by the percentage change in price

Price elasticity of supply is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Backward-bending supply curve** means a curve showing that beyond some pay level a worker offers fewer hours as pay rises further. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in exceptions to the law, before moving to the next step.

The term **Price elasticity of supply** means the percentage change in quantity supplied divided by the percentage change in price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is

being measured or compared in exceptions to the law, before moving to the next step.

The term **Elastic supply** means supply for which quantity responds by a larger percentage than price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in exceptions to the law, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in exceptions to the law to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

7.4 | ELASTICITY OF SUPPLY | TECHNIQUE | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Inelastic supply** means supply for which quantity responds by a smaller percentage than price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in elasticity of supply, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in elasticity of supply is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is

Elastic supply. supply for which quantity responds by a larger percentage than price

Elastic supply is the working idea for this part of the chapter.

Exceptions to the Law is understood by separating the question into small steps.

Exceptions to the Law is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what elasticity of supply means before applying it.

Inelastic supply. supply for which quantity responds by a smaller percentage than price

Inelastic supply is the working idea for this part of the chapter.

Elasticity of Supply is understood by separating the question into small steps.

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The section closes by restating the condition that controls the result.

examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in elasticity of supply to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

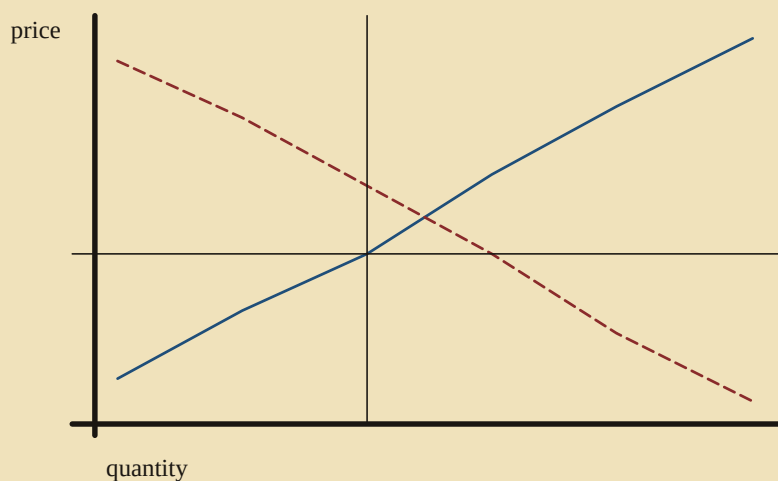


FIG. 7.1. a supply curve drawn from the printer's schedule

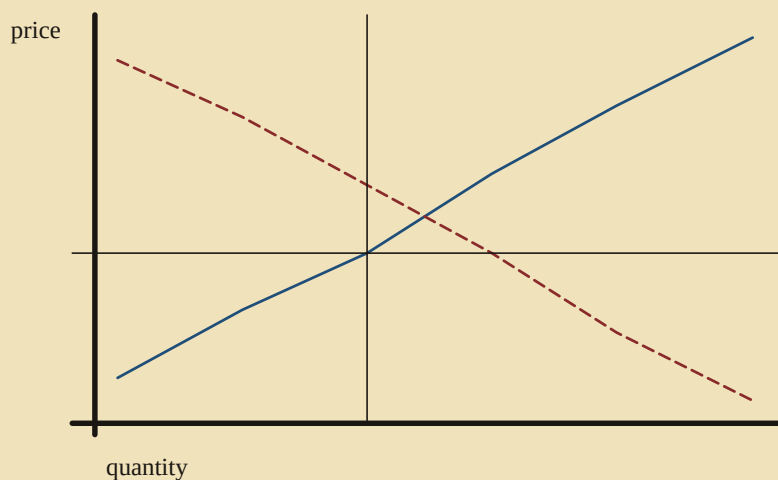


FIG. 7.2. a shift of the supply curve with its causes labelled

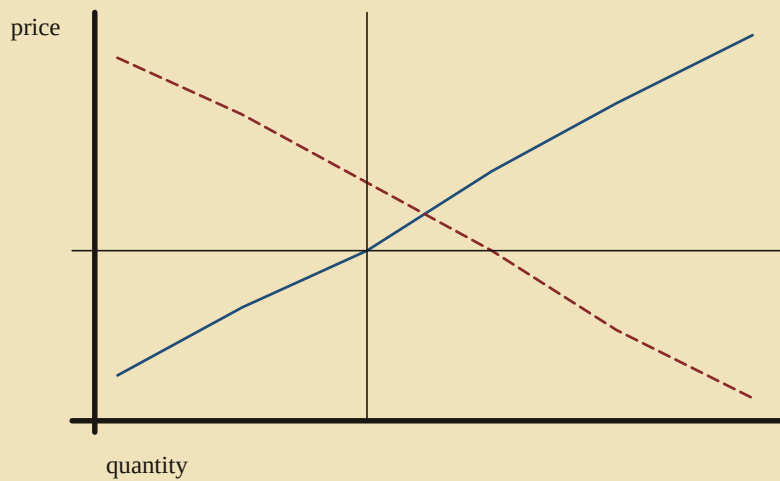


FIG. 7.3. a supply curve of labour that bends backward at high pay

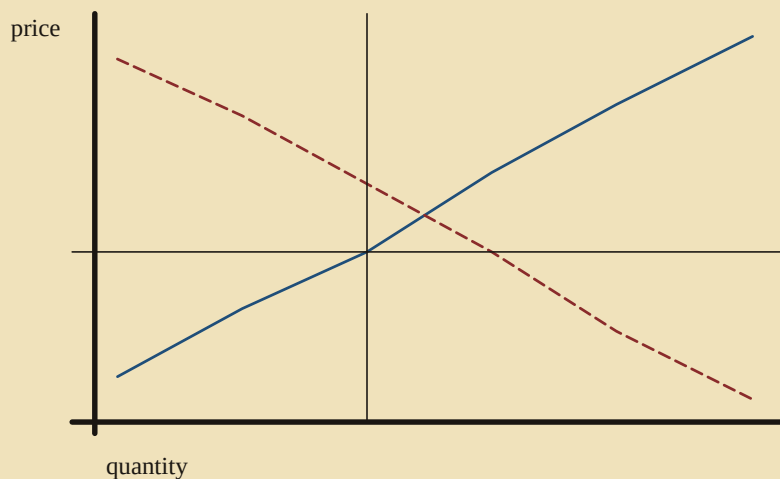


FIG. 7.4. three supply curves of different elasticity through the same point

The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

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WORKED CASE: THE RUNNING RECORD

The worked case applies the chapter without adding facts to the record.

Situation. Meghna and Sons (illustrative). A local printer supplies 200-page exercise books to the shop at a wholesale price. Weekly quantity offered, 9 March 2026: Tk 40, 150 books; Tk 45, 200; Tk 50, 250; Tk 55, 300; Tk 60, 350. Elasticity from Tk 50 to Tk 55: +10 per cent price, +20 per cent quantity, elasticity 2.0 (elastic). Shift: on 1 April 2026 the printer's paper cost rose and it offered 50 fewer books at every price: 100, 150, 200, 250, 300. Exceptional curve 1: a rare first edition, only one copy exists, supply unchanged at any price. Exceptional curve 2: a part-time binder offers 20 hours a week at Tk 150 an hour, 26 hours at Tk 200, 24 hours at Tk 260. Question. What does the record show when the chapter's central rule is applied to these facts?

The reasoning follows the data rather than an invented story.

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

The conclusion is limited to what the case supports.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

A final check asks whether the result fits the original data.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

The chapter has built one connected chain of reasoning.

Build the supply idea from a price table to a curve and measure how strongly sellers respond to price. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

7.T1 draw a supply curve from a new schedule, separate five events into movements and shifts, compute and name three elasticities of supply from new data

7.T2 explain why a farmer's supply of a perishable crop is inelastic in the short run

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

IN HISTORY

The entry directs attention to The Lancashire cotton famine of 1861 to 1865 and the rise of cotton exports from Egypt and India; search: Lancashire cotton famine 1861 Egypt India cotton exports. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The historical case shows the chapter idea in a documented setting.

REVIEW QUESTIONS

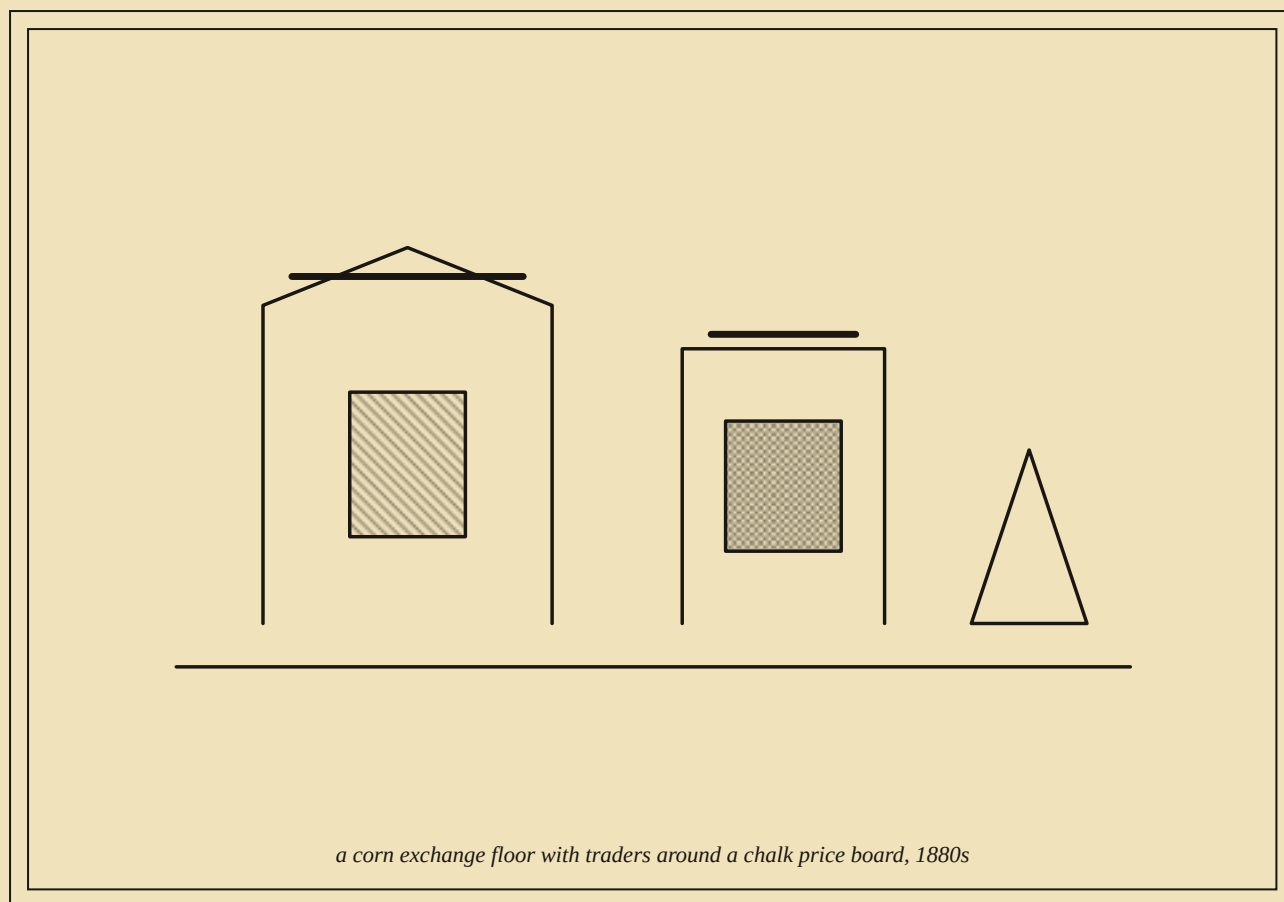
7.3 draw a supply curve from a new schedule, separate five events into movements and shifts, compute and name three elasticities of supply from new data

7.4 explain why a farmer's supply of a perishable crop is inelastic in the short run

7.5 after the shift passage and after the elasticity-of-supply formula

PAUSE AND CHECK 7.P6 after the shift passage

PAUSE AND CHECK 7.P7 the elasticity-of-supply formula



CHAPTER VIII

How Markets Find a Price

What does this chapter let us explain that the earlier ideas could not explain alone?



Bring demand and supply together to find a price, show why a market is pulled toward it, and measure who gains. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Bring demand and supply together to find a price, show why a market is pulled toward it, and measure who gains. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: Q_d 1,050, Q_s 650, excess demand 400. Floor at Tk 70: Q_d 750, Q_s 1,150, excess supply 400. Supply shift: paper cost rises, supply becomes $Q_s = 25P - 760$; new equilibrium $P = \text{Tk } 64$, $Q = 840$. Demand intercept price is Tk 120; supply intercept price is Tk 24 before the shift. Consumer surplus at equilibrium: $0.5 \times (120 - 60) \times 900 = \text{Tk } 27,000$ a week; producer surplus: $0.5 \times (60 - 24) \times 900 = \text{Tk } 16,200$ a week. After the shift: consumer surplus $0.5 \times (120 - 64) \times 840 = \text{Tk } 23,520$; producer surplus $0.5 \times (64 - 30.4) \times 840 = \text{Tk } 14,112$. Substitute-goods link: when the price of petrol rises, demand for natural gas shifts right (kept as a general example, not shop data). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

8.1 | MARKETS AND MARKET EQUILIBRIUM | FRAMEWORK | 550

This section asks what markets and market equilibrium means before applying it.

Equilibrium. a state in which nothing pushes the price or quantity to change

Equilibrium is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Equilibrium** means a state in which nothing pushes the price or quantity to change. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: Q_d 1,050, Q_s 650, excess demand 400. Floor at Tk 70: Q_d 750, Q_s 1,150, excess supply 400. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in markets and market equilibrium, before moving to the next step.

The term **Equilibrium price** means the price at which quantity demanded equals quantity supplied. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:.. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in markets and market equilibrium, before moving to the next step.

The term **Equilibrium quantity** means the quantity bought and sold at the equilibrium price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:.. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in markets and market equilibrium, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in markets and market equilibrium to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

8.2 | FORCES TOWARD EQUILIBRIUM | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Excess demand** means the amount by which quantity demanded exceeds quantity supplied at a price below equilibrium. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:.. In this chapter, the term matters because it lets the reader state precisely what is being

Equilibrium price. the price at which quantity demanded equals quantity supplied

Equilibrium price is the working idea for this part of the chapter.

Equilibrium quantity. the quantity bought and sold at the equilibrium price

Equilibrium quantity is the working idea for this part of the chapter.

Markets and Market Equilibrium is understood by separating the question into small steps.

Markets and Market Equilibrium is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what forces toward equilibrium means before applying it.

Excess demand. the amount by which quantity demanded exceeds quantity supplied at a price below equilibrium

Excess demand is the working idea for this part of the chapter.

Excess supply. the amount by which quantity supplied exceeds quantity demanded at a price above equilibrium

Excess supply is the working idea for this part of the chapter.

Price ceiling. a legal maximum price that may be charged for a good

Price ceiling is the working idea for this part of the chapter.

Forces Toward Equilibrium is understood by separating the question into small steps.

Forces Toward Equilibrium is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what shifts and new equilibria means before applying it.

Price floor. a legal minimum price that may be charged for a good

Price floor is the working idea for this part of the chapter.

measured or compared in forces toward equilibrium, before moving to the next step.

The term **Excess supply** means the amount by which quantity supplied exceeds quantity demanded at a price above equilibrium. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forces toward equilibrium, before moving to the next step.

The term **Price ceiling** means a legal maximum price that may be charged for a good. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk 70:. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in forces toward equilibrium, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in forces toward equilibrium to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

8.3 | SHIFTS AND NEW EQUILIBRIA | TECHNIQUE | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Price floor** means a legal minimum price that may be charged for a good. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: $Q_d 1,050$, $Q_s 650$, excess demand 400. Floor at Tk

70:.. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in shifts and new equilibria, before moving to the next step.

The term **Producer surplus** means the gap between the price a seller receives and the lowest price the seller would accept. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: Q_d 1,050, Q_s 650, excess demand 400. Floor at Tk 70:.. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in shifts and new equilibria, before moving to the next step.

The term **Total surplus** means consumer surplus plus producer surplus, a measure of the gain from trade. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Neighbourhood market for 200-page exercise books, weekly: demand $Q_d = 1,800 - 15P$ and supply $Q_s = 25P - 600$, with P in taka. Equilibrium: $P = \text{Tk } 60$, $Q = 900$. Ceiling at Tk 50: Q_d 1,050, Q_s 650, excess demand 400. Floor at Tk 70:.. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in shifts and new equilibria, before moving to the next step.

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The central question in shifts and new equilibria is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in shifts and new equilibria to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

8.4 | CONSUMER AND PRODUCER SURPLUS IN PRACTICE | NEW IDEA | 450

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the

Producer surplus. the gap between the price a seller receives and the lowest price the seller would accept

Producer surplus is the working idea for this part of the chapter.

Total surplus. consumer surplus plus producer surplus, a measure of the gain from trade

Total surplus is the working idea for this part of the chapter.

Shifts and New Equilibria is understood by separating the question into small steps.

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The section closes by restating the condition that controls the result.

This section asks what consumer and producer surplus in practice means before applying it.

Consumer and Producer Surplus in Practice is understood by separating the question into small steps.

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The central question in consumer and producer surplus in practice is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

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Before moving on, keep one question in view: what must remain true for the rule in consumer and producer surplus in practice to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

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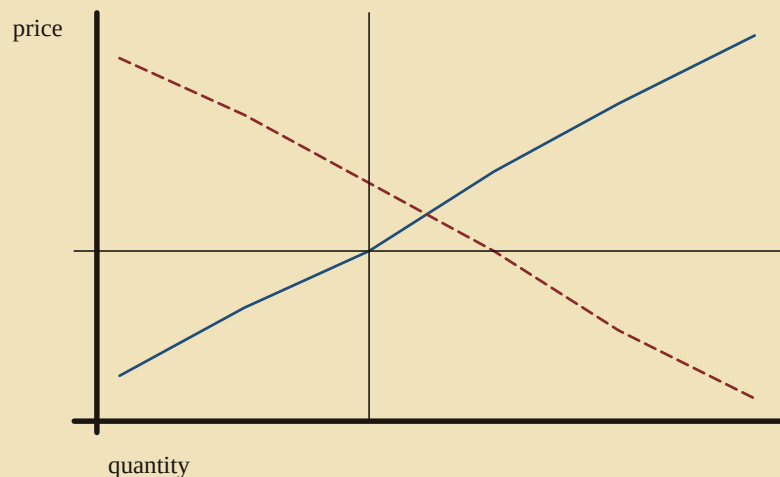


FIG. 8.1. demand and supply curves crossing at the equilibrium point

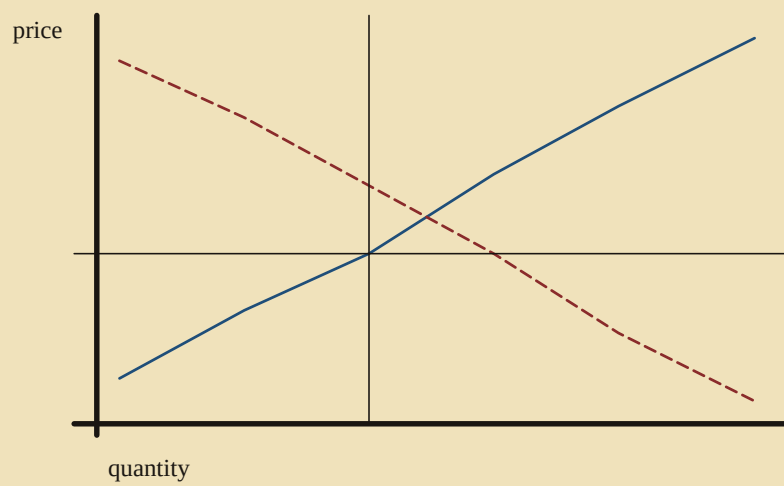


FIG. 8.2. a ceiling below and a floor above equilibrium with the shortage and surplus gaps marked

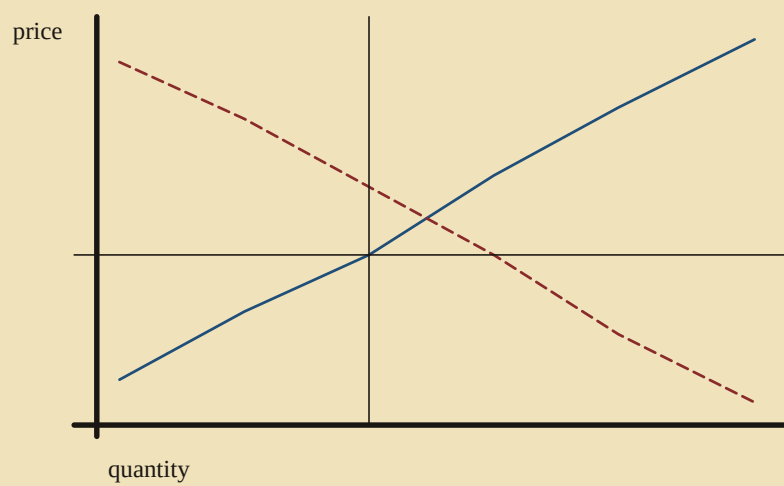


FIG. 8.3. two panels: a supply shift left and a demand shift right with the new equilibria

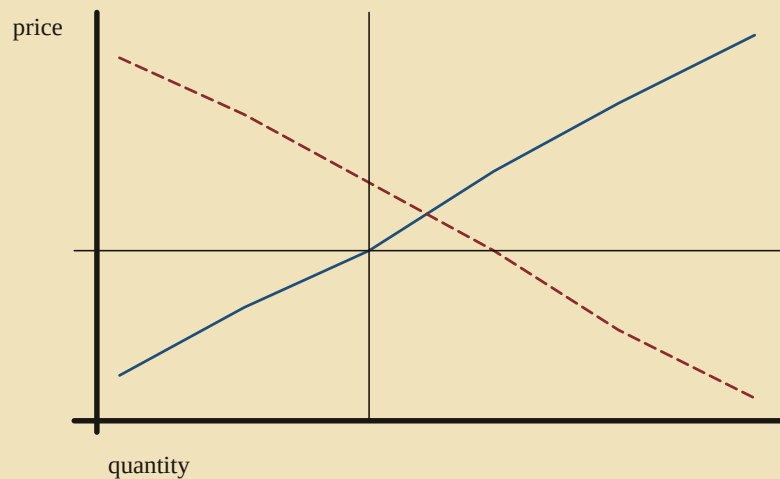


FIG. 8.4. consumer surplus and producer surplus shaded on one diagram

The same method remains useful when the surrounding details change.

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The reader should separate the starting condition from the change being examined, then identify the quantity that responds. This prevents a movement from being confused with a shift, a total from being confused with a marginal amount, or a descriptive fact from being confused with a judgement. The running case provides fixed figures, but the reasoning is broader: identify the relevant definition, state the condition, apply the relationship, and check the result against the original data. If one condition changes, the analysis must be repeated rather than carried over unchanged. This disciplined sequence is the main practical skill developed by the chapter.

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Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

The reasoning follows the data rather than an invented story.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

The conclusion is limited to what the case supports.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

A final check asks whether the result fits the original data.

* * *

SUMMARY

Bring demand and supply together to find a price, show why a market is pulled toward it, and measure who gains. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

The chapter has built one connected chain of reasoning.

THINK IT THROUGH

8.T1 find equilibrium from new equations, describe the adjustment from a price above or below it, trace the effect of a shift

8.T2 a government sets a price ceiling below equilibrium for a staple — describe who gains and who loses

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

A common error is to apply a rule after its condition has changed.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

Another error is to confuse a numerical result with the meaning of that result.

IN HISTORY

The entry directs attention to Diocletian's Edict on Maximum Prices, AD 301; search: Diocletian Edict on Maximum Prices 301 shortages. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The historical case shows the chapter idea in a documented setting.

REVIEW QUESTIONS

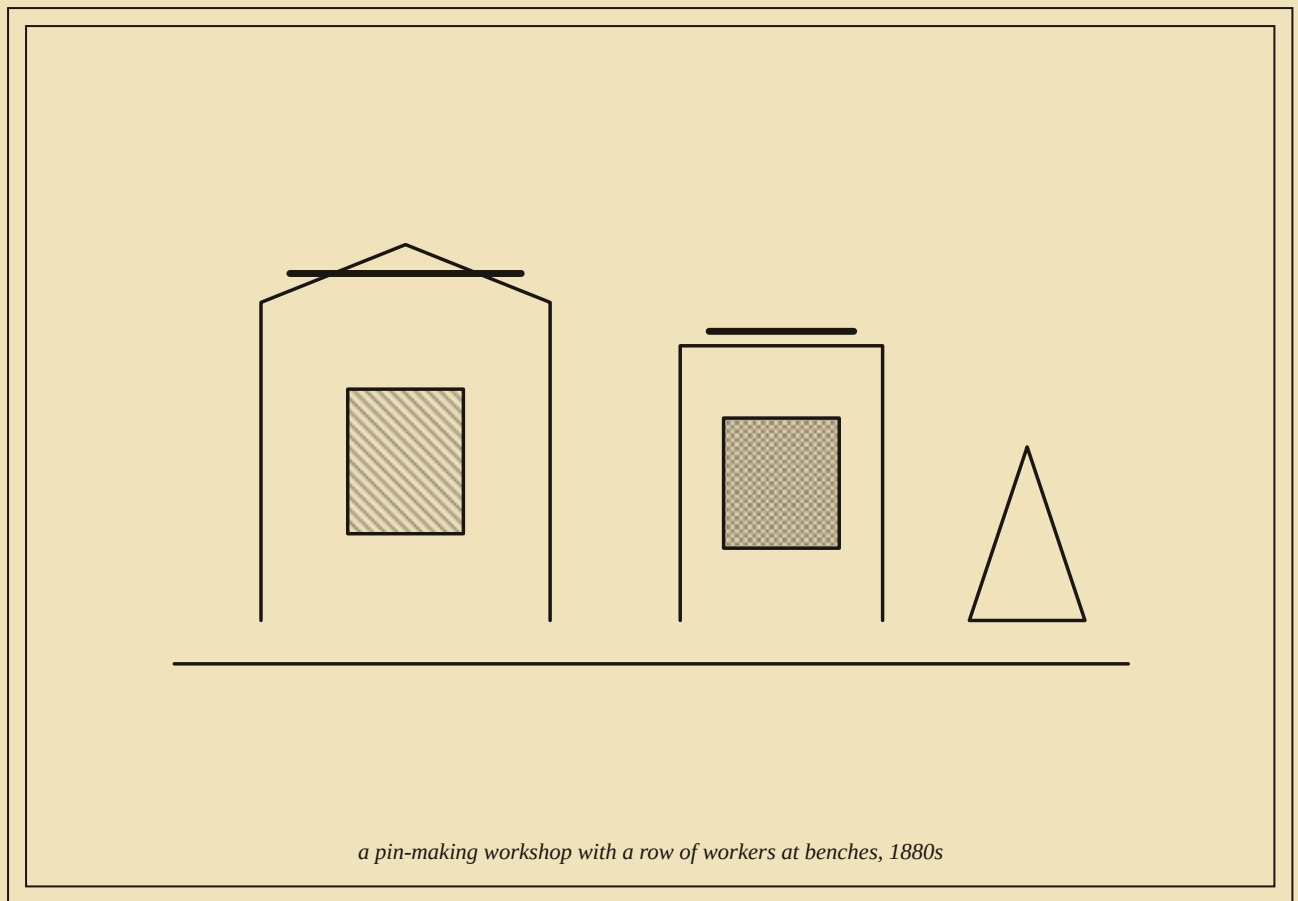
8.3 find equilibrium from new equations, describe the adjustment from a price above or below it, trace the effect of a shift

8.4 a government sets a price ceiling below equilibrium for a staple — describe who gains and who loses

8.5 after the equilibrium solution and after the shortage-and-surplus passage

PAUSE AND CHECK 8.P6 after the equilibrium solution

PAUSE AND CHECK 8.P7 the shortage-and-surplus passage



CHAPTER IX

Production

What does this chapter let us explain that the earlier ideas could not explain alone?



Explain how inputs become output, why adding one input yields less and less, and what changes when all inputs grow. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Explain how inputs become output, why adding one input yields less and less, and what changes when all inputs grow. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12, 12, 11, 9.6, 8. Long run, scale tables (binds a day): one machine and 2 workers, 40; two machines and 4 workers, 90; four machines and 8 workers, 180; eight machines and 16 workers, 300. Types: first doubling gives 2.25 times output (increasing), second doubling gives 2.0 times (constant), third gives about 1.67 times (decreasing). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

9.1 | THE PRODUCTION FUNCTION | NEW IDEA | 650

This section asks what the production function means before applying it.

Production. the process of turning resources into goods and services

Production is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Production** means the process of turning resources into goods and services. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the production function, before moving to the next step.

Input. a resource used in production

Input is the working idea for this part of the chapter.

The term **Input** means a resource used in production. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14,

12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the production function, before moving to the next step.

The term **Output** means the goods or services produced. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the production function, before moving to the next step.

The term **Production function** means a rule or table linking the quantities of inputs used to the greatest output they can make. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in the production function, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in the production function to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

9.2 | SHORT RUN AND LONG RUN | FRAMEWORK |

225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

Output. the goods or services produced

Output is the working idea for this part of the chapter.

Production function. a rule or table linking the quantities of inputs used to the greatest output they can make

Production function is the working idea for this part of the chapter.

The Production Function is understood by separating the question into small steps.

The Production Function is understood by separating the question into small steps.

The Production Function is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what short run and long run means before applying it.

Short run. a period too short to change at least one input, such as machines or floor space

Short run is the working idea for this part of the chapter.

Long run. a period long enough to change every input

Long run is the working idea for this part of the chapter.

Fixed input. an input whose quantity cannot be changed in the short run

Fixed input is the working idea for this part of the chapter.

Variable input. an input whose quantity can be changed even in the short run

Variable input is the working idea for this part of the chapter.

Short Run and Long Run is understood by separating the question into small steps.

Short Run and Long Run is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The term **Short run** means a period too short to change at least one input, such as machines or floor space. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short run and long run, before moving to the next step.

The term **Long run** means a period long enough to change every input. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short run and long run, before moving to the next step.

The term **Fixed input** means an input whose quantity cannot be changed in the short run. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short run and long run, before moving to the next step.

The term **Variable input** means an input whose quantity can be changed even in the short run. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short run and long run, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in short run and long run to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

9.3 | DIMINISHING RETURNS | DEFINING THEORY | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Total product** means the whole output from a given amount of inputs. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in diminishing returns, before moving to the next step.

The term **Average product** means total product divided by the number of units of the variable input. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in diminishing returns, before moving to the next step.

The term **Marginal product** means the extra output from adding one more unit of the variable input. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in diminishing returns, before moving to the next step.

The term **Law of diminishing returns** means when more of one input is added to fixed inputs, the extra output eventually falls. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in diminishing returns, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

This section asks what diminishing returns means before applying it.

Total product. the whole output from a given amount of inputs

Total product is the working idea for this part of the chapter.

Average product. total product divided by the number of units of the variable input

Average product is the working idea for this part of the chapter.

Marginal product. the extra output from adding one more unit of the variable input

Marginal product is the working idea for this part of the chapter.

Law of diminishing returns. when more of one input is added to fixed inputs, the extra output eventually falls

Law of diminishing returns is the working idea for this part of the chapter.

Diminishing Returns is understood by separating the question into small steps.

Diminishing Returns is understood by separating the question into small steps.

Diminishing Returns is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in diminishing returns is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in diminishing returns to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

9.4 | RETURNS TO SCALE | TECHNIQUE | 450

This section asks what returns to scale means before applying it.

Returns to scale. the change in output when all inputs are changed by the same proportion

Returns to scale is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Returns to scale** means the change in output when all inputs are changed by the same proportion. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in returns to scale, before moving to the next step.

Increasing returns to scale. output rises by a larger proportion than inputs

Increasing returns to scale is the working idea for this part of the chapter.

The term **Increasing returns to scale** means output rises by a larger proportion than inputs. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in returns to scale, before moving to the next step.

Constant returns to scale. output rises by the same proportion as inputs

Constant returns to scale is the working idea for this part of the chapter.

The term **Constant returns to scale** means output rises by the same proportion as inputs. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in returns to scale, before moving to the next step.

The term **Decreasing returns to scale** means output rises by a smaller proportion than inputs. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in returns to scale, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in returns to scale is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in returns to scale to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

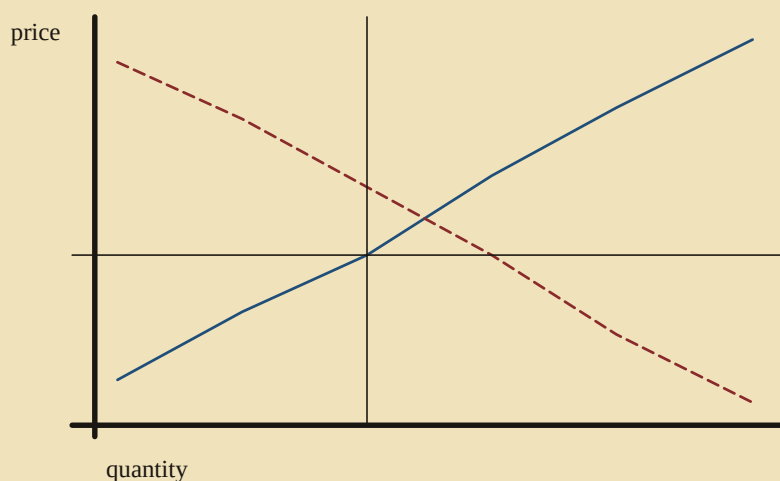


FIG. 9.1. a total product curve rising then flattening and a marginal product curve peaking then falling

Decreasing returns to scale.

output rises by a smaller proportion than inputs

Decreasing returns to scale is the working idea for this part of the chapter.

Returns to Scale is understood by separating the question into small steps.

Returns to Scale is understood by separating the question into small steps.

Returns to Scale is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

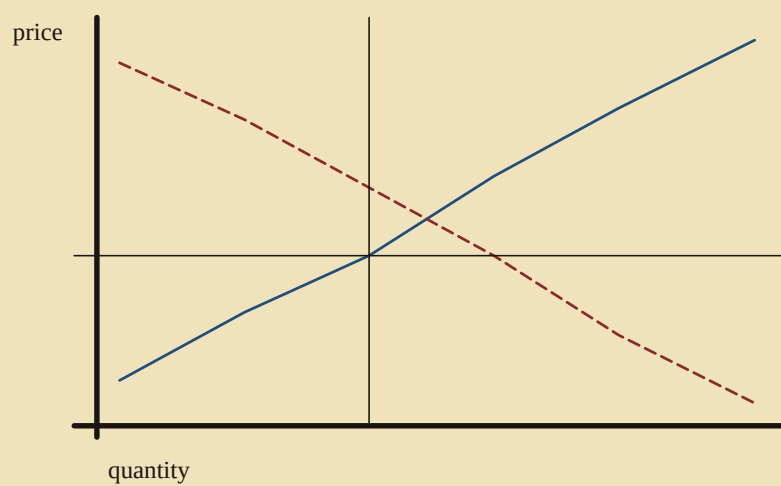


FIG. 9.2. *average and marginal product on one diagram showing where they meet*

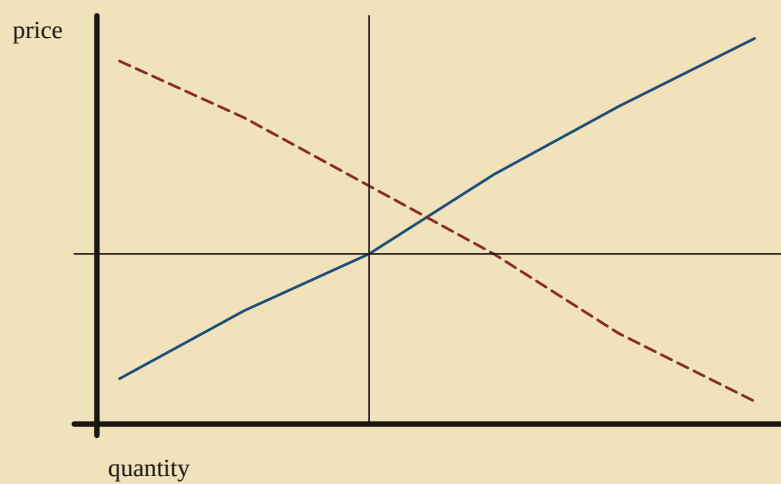


FIG. 9.3. *a time bar marking the short run and the long run*

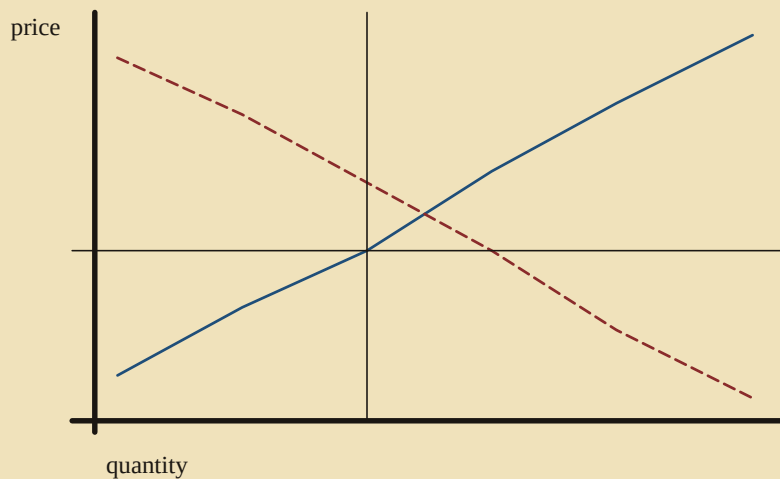


FIG. 9.4. three paths of output as scale doubles: more than double, double, less than double

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). On 3 March 2026 the shop opened a binding corner for theses with one binding machine. Short run, one machine and 1 to 6 workers, binds a day: 10, 24, 36, 44, 48, 48. Marginal product: 10, 14, 12, 8, 4, 0. Average product: 10, 12, 12, 11, 9.6, 8. Long run, scale tables (binds a day): one machine and 2 workers, 40; two machines and 4 workers, 90; four machines and 8 workers, 180; eight machines and 16 workers, 300. Types: first doubling gives 2.25 times output (increasing), second doubling gives 2.0 times (constant), third gives about 1.67 times (decreasing). Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Explain how inputs become output, why adding one input yields less and less, and what changes when all inputs grow. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been

The worked case applies the chapter without adding facts to the record.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

9.T1 complete average and marginal columns from new data, say where diminishing returns start, classify the returns to scale in a new table

9.T2 explain why a factory cannot directly add workers forever to raise output

WHERE PEOPLE GO WRONG

A common error is to apply a rule after its condition has changed.

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Another error is to confuse a numerical result with the meaning of that result.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The historical case shows the chapter idea in a documented setting.

The entry directs attention to Henry Ford's moving assembly line at Highland Park, 1913, and the fall in chassis assembly time; search: Ford moving assembly line 1913 chassis assembly time hours. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

REVIEW QUESTIONS

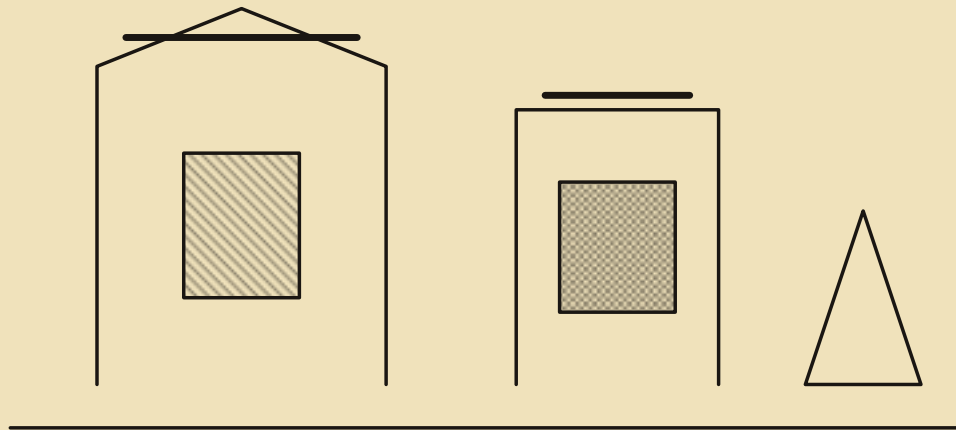
9.3 complete average and marginal columns from new data, say where diminishing returns start, classify the returns to scale in a new table

9.4 explain why a factory cannot directly add workers forever to raise output

9.5 after the short-run versus long-run passage and after the marginal-product table

PAUSE AND CHECK 9.P6 after the short-run versus long-run passage

PAUSE AND CHECK 9.P7 the marginal-product table



a foundry floor with a tilting converter and ladles of molten metal, 1880s

CHAPTER X

Costs of Production

What does this chapter let us explain that the earlier ideas could not explain alone?



Show what costs a firm counts, how they change with output, and how to reach a given output at the lowest cost. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Show what costs a firm counts, how they change with output, and how to reach a given output at the lowest cost. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output (binds a month) 100, 200, 300, 400, 500, 600, 700 with variable cost Tk 4,000; 7,000; 9,000; 12,000; 17,000; 24,000; 34,000. Total cost Tk 24,000; 27,000; 29,000; 32,000; 37,000; 44,000; 54,000. Average cost Tk 240; 135; 96.67; 80; 74; 73.33; 77.14. Marginal cost per bind between steps: Tk 40; 30; 20; 30; 50; 70; 100 (marginal cost crosses average cost near 600 binds, where average cost is lowest at Tk 73.33). Long run: with one machine lowest average cost Tk 73.33 at 600 binds; with two machines Tk 66 at 1,200; with three machines Tk 70 at 1,800. Least-cost combination for 400 binds a day, wage Tk 600 a day and machine rent Tk 1,200 a day: 10 workers and 5 machines cost Tk 12,000; 16 workers and 3 machines Tk 13,200; 8 workers and 8 machines Tk 14,400; 24 workers and 2 machines Tk 16,800; least cost Tk 12,000, where marginal product of labour is 30 and of a machine 60 (30 over 600 equals 60 over 1,200). The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

10.1 | COST CONCEPTS | FRAMEWORK | 450

This section asks what cost concepts means before applying it.

Cost. the value of the resources used up in making a good or service

Cost is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Cost** means the value of the resources used up in making a good or service. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely

what is being measured or compared in cost concepts, before moving to the next step.

The term **Explicit cost** means a cost paid out in money to others, such as rent or wages. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in cost concepts, before moving to the next step.

The term **Implicit cost** means the value of resources the owner supplies, such as own time, for which no payment is made. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in cost concepts, before moving to the next step.

The term **Fixed cost** means a cost that stays the same however much is produced in the short run. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in cost concepts, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

Before moving on, keep one question in view: what must remain true for the rule in cost concepts to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

Explicit cost. a cost paid out in money to others, such as rent or wages

Explicit cost is the working idea for this part of the chapter.

Implicit cost. the value of resources the owner supplies, such as own time, for which no payment is made

Implicit cost is the working idea for this part of the chapter.

Fixed cost. a cost that stays the same however much is produced in the short run

Fixed cost is the working idea for this part of the chapter.

Cost Concepts is understood by separating the question into small steps.

Cost Concepts is understood by separating the question into small steps.

Cost Concepts is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

10.2 | SHORT-RUN COST CURVES | TECHNIQUE | 775

This section asks what short-run cost curves means before applying it.

Variable cost. a cost that rises and falls with the amount produced

Variable cost is the working idea for this part of the chapter.

Total cost. fixed cost plus variable cost at a given output

Total cost is the working idea for this part of the chapter.

Average cost. total cost divided by the number of units produced

Average cost is the working idea for this part of the chapter.

Marginal cost. the extra cost of producing one more unit

Marginal cost is the working idea for this part of the chapter.

Short-Run Cost Curves is understood by separating the question into small steps.

Short-Run Cost Curves is understood by separating the question into small steps.

Short-Run Cost Curves is understood by separating the question into small steps.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Variable cost** means a cost that rises and falls with the amount produced. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short-run cost curves, before moving to the next step.

The term **Total cost** means fixed cost plus variable cost at a given output. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short-run cost curves, before moving to the next step.

The term **Average cost** means total cost divided by the number of units produced. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short-run cost curves, before moving to the next step.

The term **Marginal cost** means the extra cost of producing one more unit. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in short-run cost curves, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a

condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in short-run cost curves to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

10.3 | LONG-RUN COST CURVES | NEW IDEA | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Prime cost** means the direct cost of materials and labour that go into a unit of output. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The term **Supplementary cost** means the cost of items apart from prime cost, such as rent and equipment, which remain even when output is nil. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The term **Long-run average cost curve** means a curve showing the lowest average cost of each output when every input can be changed. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The term **Isoquant** means a curve joining the input combinations that all make the same output. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost.

The section closes by restating the condition that controls the result.

This section asks what long-run cost curves means before applying it.

Prime cost. the direct cost of materials and labour that go into a unit of output

Prime cost is the working idea for this part of the chapter.

Supplementary cost. the cost of items apart from prime cost, such as rent and equipment, which remain even when output is nil

Supplementary cost is the working idea for this part of the chapter.

Long-run average cost curve. a curve showing the lowest average cost of each output when every input can be changed

Long-run average cost curve is the working idea for this part of the chapter.

Isoquant. a curve joining the input combinations that all make the same output

Isoquant is the working idea for this part of the chapter.

Long-Run Cost Curves is understood by separating the question into small steps.

Long-Run Cost Curves is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what least-cost combination means before applying it.

Isocost line. a line joining the input combinations that all cost the same total

Isocost line is the working idea for this part of the chapter.

Least-cost combination. the mix of inputs that makes a given output for the lowest total cost

Least-cost combination is the working idea for this part of the chapter.

Least-Cost Combination is understood by separating the question into small steps.

Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in long-run cost curves, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in long-run cost curves to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

10.4 | LEAST-COST COMBINATION | TECHNIQUE | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Isocost line** means a line joining the input combinations that all cost the same total. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in least-cost combination, before moving to the next step.

The term **Least-cost combination** means the mix of inputs that makes a given output for the lowest total cost. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in least-cost combination, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in least-cost combination is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in least-cost combination to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

Least-Cost Combination is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

* * *

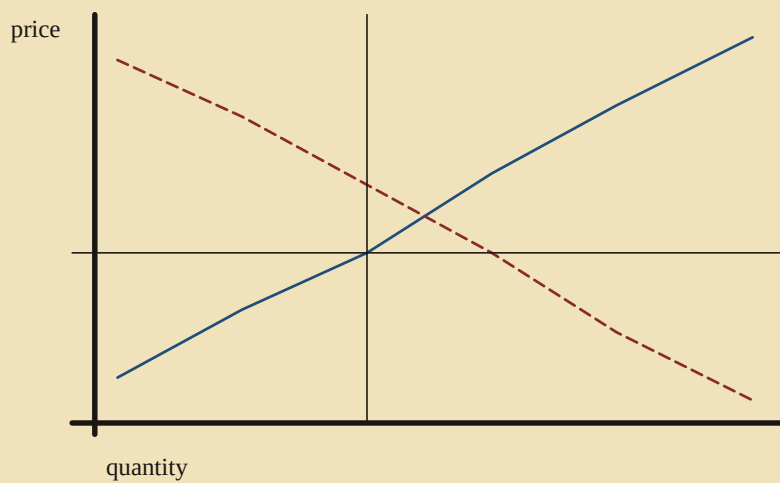


FIG. 10.1. *fixed, variable and total cost curves on one diagram*

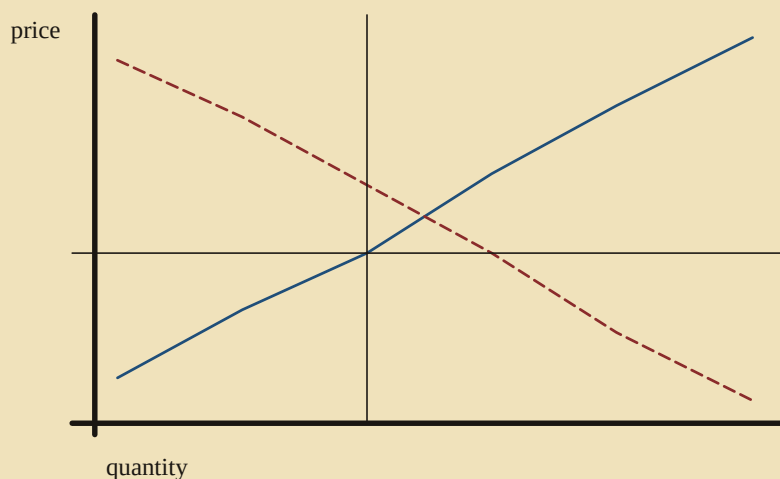


FIG. 10.2. *average cost and marginal cost curves, with marginal cost cutting average cost at its lowest point*

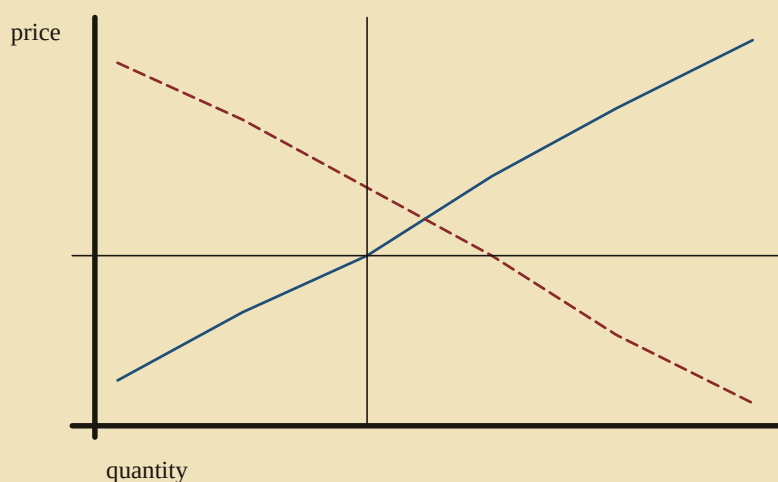


FIG. 10.3. *three short-run average cost curves with the long-run curve enveloping them*

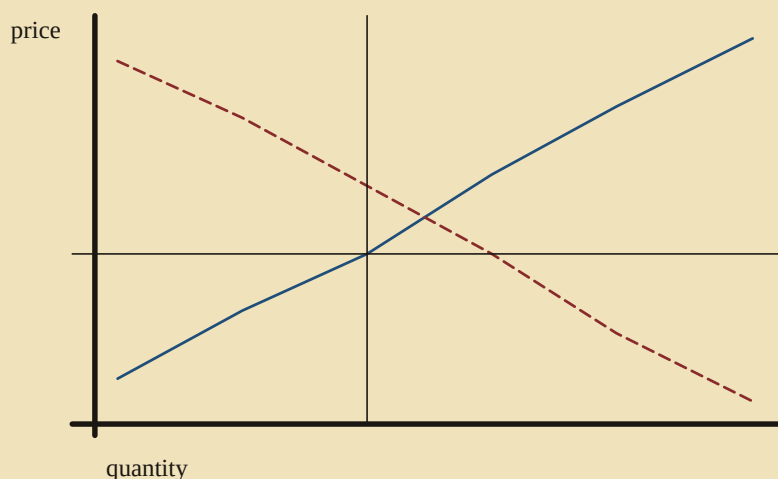


FIG. 10.4. *an isocost line touching an isoquant at the least-cost point*

WORKED CASE: THE RUNNING RECORD

The worked case applies the chapter without adding facts to the record.

Situation. Meghna and Sons (illustrative). Binding corner monthly, 2026. Fixed cost Tk 20,000 (corner rent Tk 12,000 plus machine instalment Tk 8,000). Implicit cost: owners' unpaid time Tk 15,000 a month. Prime cost is materials and labour, equal to variable cost. Output (binds a month) 100, 200, 300, 400, 500, 600, 700 with variable cost Tk 4,000; 7,000; 9,000; 12,000; 17,000; 24,000; 34,000. Total cost Tk 24,000; 27,000; 29,000; 32,000; 37,000; 44,000; 54,000. Average cost Tk 240; 135; 96.67; 80; 74; 73.33; 77.14. Marginal cost per bind between steps: Tk 40; 30; 20; 30; 50; 70; 100 (marginal cost crosses average cost near 600 binds, where average cost is lowest at Tk 73.33). Long run: with one machine lowest average cost Tk 73.33 at 600 binds; with two machines Tk 66 at 1,200; with three machines Tk 70 at 1,800. Least-cost combination for 400 binds a day, wage Tk 600 a day and machine rent Tk 1,200 a day: 10 workers and 5 machines cost Tk 12,000; 16 workers and 3 machines Tk 13,200; 8

workers and 8 machines Tk 14,400; 24 workers and 2 machines Tk 16,800; least cost Tk 12,000, where marginal product of labour is 30 and of a machine 60 (30 over 600 equals 60 over 1,200). Question. What does the record show when the chapter's central rule is applied to these facts?

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Show what costs a firm counts, how they change with output, and how to reach a given output at the lowest cost. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

10.T1 complete the cost columns from new data, find where marginal cost crosses average cost, choose the cheapest input mix from three priced options

10.T2 explain why a family business should count its own unpaid time as a cost when judging success

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to Henry Bessemer's converter process, 1856, and the fall in the price of steel; search: Bessemer process 1856 steel price fall per ton. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

The reasoning follows the data rather than an invented story.

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

REVIEW QUESTIONS

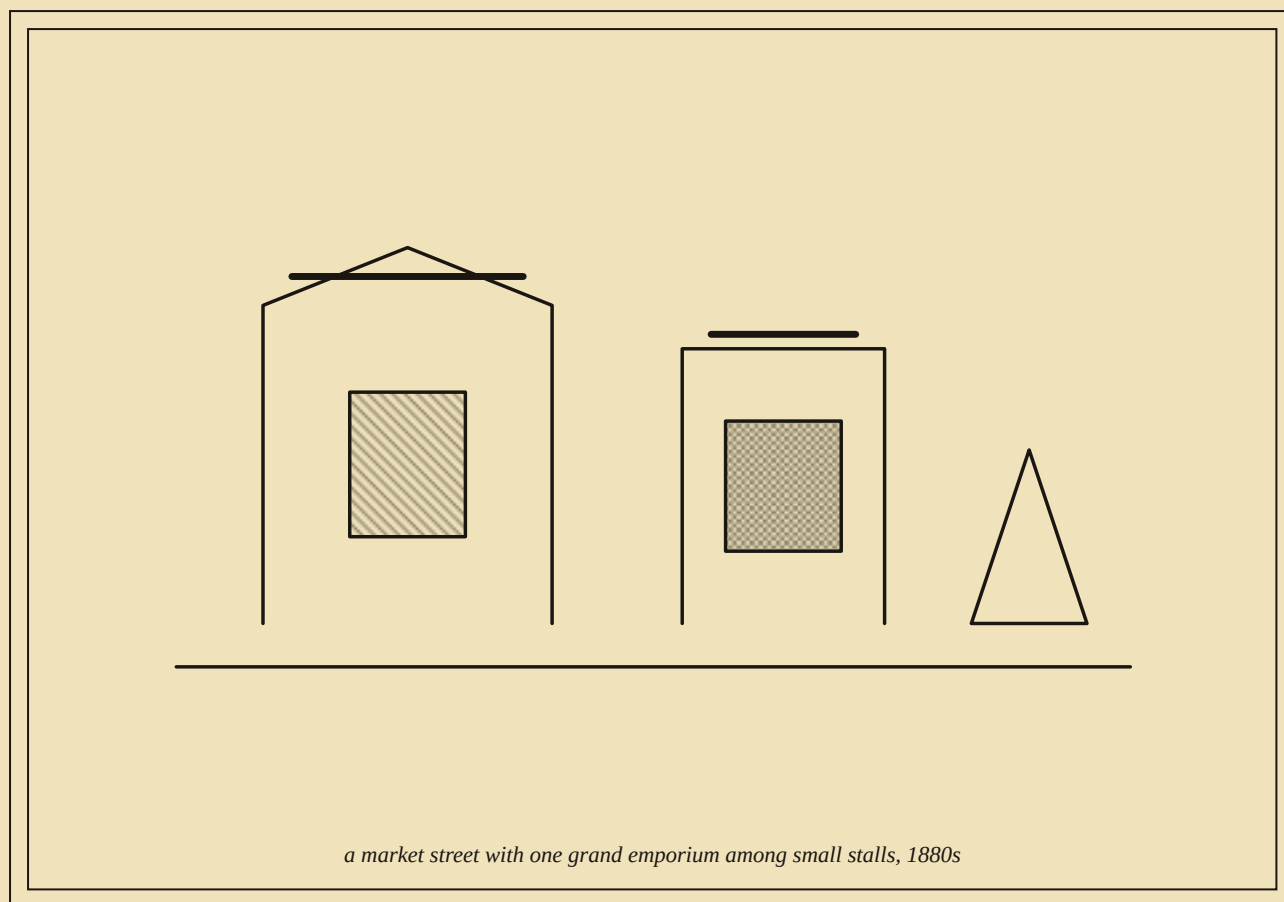
10.3 complete the cost columns from new data, find where marginal cost crosses average cost, choose the cheapest input mix from three priced options

10.4 explain why a family business should count its own unpaid time as a cost when judging success

10.5 after the average-versus-marginal passage and after the least-cost rule

PAUSE AND CHECK 10.P6 after the average-versus-marginal passage

PAUSE AND CHECK 10.P7 the least-cost rule



CHAPTER XI

Market Structures and the Firm

What does this chapter let us explain that the earlier ideas could not explain alone?



Compare how firms behave when they face many rivals, one rival or none, and how each structure sets price and output. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Compare how firms behave when they face many rivals, one rival or none, and how each structure sets price and output. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70; 50, Tk 80. Profit-maximising output 30 (marginal cost equals Tk 60). Average cost at 30 books Tk 52; profit $(60 - 52) \times 30 = \text{Tk } 240$ a day. Entry of new shops lowers price toward average cost in the long run. Monopoly: sole authorised seller of one foreign reference work. Price and weekly quantity: Tk 1,000, 10 copies; Tk 900, 20; Tk 800, 30; Tk 700, 40. Total revenue Tk 10,000; 18,000; 24,000; 28,000. Marginal revenue between steps Tk 800; 600; 400. Marginal cost constant Tk 500. Profit-maximising output 30 copies at Tk 800; profit $(800 - 500) \times 30 = \text{Tk } 9,000$. Monopolistic competition: a mystery novel series differentiated by cover and author; price Tk 400; short run average cost Tk 340 at 50 copies a week; profit Tk 3,000. After two more shops enter, the shop sells 40 copies at Tk 380 with average cost Tk 380; profit nil. Pricing: price cut from Tk 60 to Tk 55 for the exercise book raised weekly sales from 200 to 210; revenue Tk 12,000 fell to Tk 11,550 (inelastic). Market structure ranking for the shop: exercise books near perfect competition; the sole-authorised reference work a monopoly; mystery novels monopolistic competition. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

11.1 | TYPES OF MARKET | FRAMEWORK | 225

This section asks what types of market means before applying it.

Market structure. the features of a market, such as number of sellers, type of product and ease of entry, that shape how firms behave
Market structure is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Market structure** means the features of a market, such as number of sellers, type of product and ease of entry, that shape how firms behave. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10

books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in types of market, before moving to the next step.

The term **Competitive market** means a market with many buyers and sellers, so no single one can set the price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in types of market, before moving to the next step.

The term **Industry** means all the firms that make the same or very similar goods. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in types of market, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in types of market to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

11.2 | PERFECT COMPETITION | NEW IDEA | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Price taker** means a firm that must accept the market price because it is too small to change it. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in perfect competition, before moving to the next step.

Competitive market. a market with many buyers and sellers, so no single one can set the price

Competitive market is the working idea for this part of the chapter.

Industry. all the firms that make the same or very similar goods

Industry is the working idea for this part of the chapter.

Types of Market is understood by separating the question into small steps.

Types of Market is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what perfect competition means before applying it.

Price taker. a firm that must accept the market price because it is too small to change it

Price taker is the working idea for this part of the chapter.

Price maker. a firm large enough that its own output choices affect the price

Price maker is the working idea for this part of the chapter.

Marginal revenue. the extra revenue gained by selling one more unit

Marginal revenue is the working idea for this part of the chapter.

Perfect Competition is understood by separating the question into small steps.

Perfect Competition is understood by separating the question into small steps.

Perfect Competition is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

The term **Price maker** means a firm large enough that its own output choices affect the price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in perfect competition, before moving to the next step.

The term **Marginal revenue** means the extra revenue gained by selling one more unit. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in perfect competition, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in perfect competition to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

11.3 | MONOPOLY | NEW IDEA | 550

This section asks what monopoly means before applying it.

Perfect competition. a market with many sellers, an identical product, free entry and full information, so each firm takes the price

Perfect competition is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Perfect competition** means a market with many sellers, an identical product, free entry and full information, so each firm takes the price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely

what is being measured or compared in monopoly, before moving to the next step.

The term **Pure competition** means a market with many sellers, an identical product and free entry, where buyers may lack full information or free movement. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopoly, before moving to the next step.

The term **Monopoly** means a market with one seller and no close substitute, so that seller sets the price. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopoly, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in monopoly to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

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11.4 | MONOPOLISTIC COMPETITION | NEW IDEA | 550

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Pure monopoly** means a monopoly in which one seller supplies the whole market with a product that has no close substitute. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopolistic competition, before moving to the next step.

Pure competition. a market with many sellers, an identical product and free entry, where buyers may lack full information or free movement

Pure competition is the working idea for this part of the chapter.

Monopoly. a market with one seller and no close substitute, so that seller sets the price

Monopoly is the working idea for this part of the chapter.

Monopoly is understood by separating the question into small steps.

Monopoly is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what monopolistic competition means before applying it.

Pure monopoly. a monopoly in which one seller supplies the whole market with a product that has no close substitute

Pure monopoly is the working idea for this part of the chapter.

Bilateral monopoly. a market with a single seller facing a single buyer

Bilateral monopoly is the working idea for this part of the chapter.

Monopolistic competition. a market with many sellers of similar but not identical goods and free entry

Monopolistic competition is the working idea for this part of the chapter.

Monopolistic Competition is understood by separating the question into small steps.

Monopolistic Competition is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what pricing and the firm's goal means before applying it.

Product differentiation. making a firm's good different from rivals' in quality, design, brand or service

Product differentiation is the working idea for this part of the chapter.

The term **Bilateral monopoly** means a market with a single seller facing a single buyer. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopolistic competition, before moving to the next step.

The term **Monopolistic competition** means a market with many sellers of similar but not identical goods and free entry. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in monopolistic competition, before moving to the next step.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in monopolistic competition is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

Before moving on, keep one question in view: what must remain true for the rule in monopolistic competition to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

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11.5 | PRICING AND THE FIRM'S GOAL | FRAMEWORK | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Product differentiation** means making a firm's good different from rivals' in quality, design, brand or service. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in pricing and the firm's goal, before moving to the next step.

The term **Oligopoly** means a market with a few large sellers, each aware of the others' actions. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in pricing and the firm's goal, before moving to the next step.

The term **Profit maximisation** means choosing the output at which the gap between total revenue and total cost is greatest. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in pricing and the firm's goal, before moving to the next step.

The central question in pricing and the firm's goal is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in pricing and the firm's goal to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

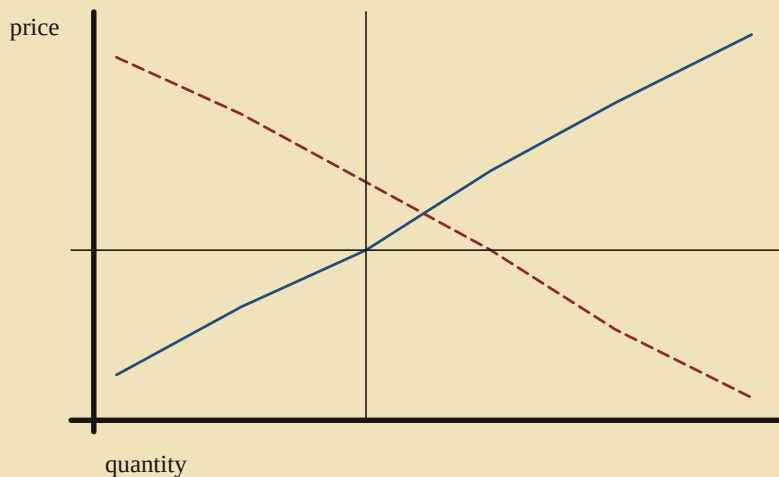


FIG. 11.1. a table-diagram of the market structures by number of sellers, product type and entry

Oligopoly. a market with a few large sellers, each aware of the others' actions

Oligopoly is the working idea for this part of the chapter.

Profit maximisation. choosing the output at which the gap between total revenue and total cost is greatest

Profit maximisation is the working idea for this part of the chapter.

Pricing and the Firm's Goal is understood by separating the question into small steps.

Pricing and the Firm's Goal is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

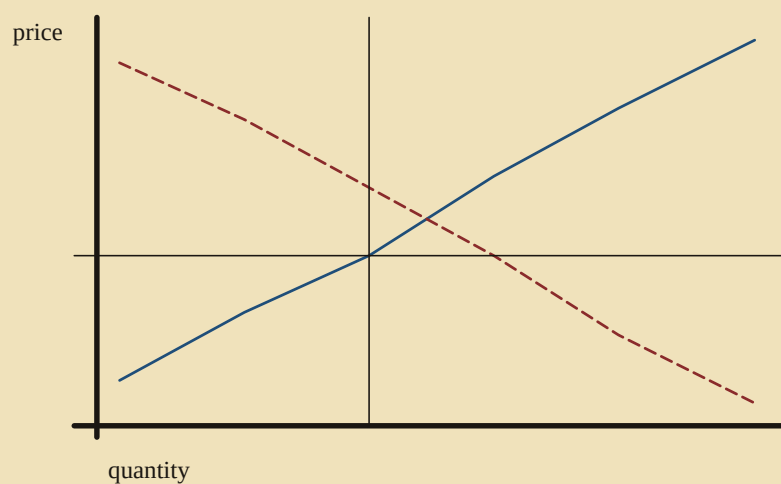


FIG. 11.2. *a perfectly competitive firm at the point where marginal cost equals price, with industry beside it*

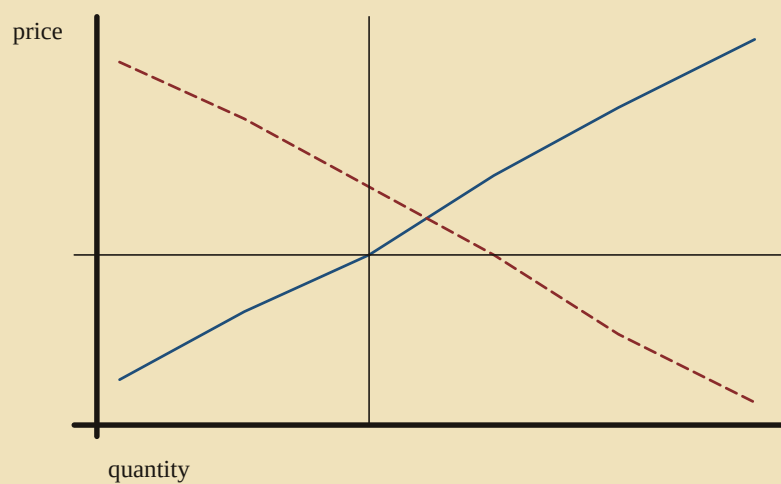


FIG. 11.3. *a monopolist's demand and marginal revenue curves with the profit rectangle*

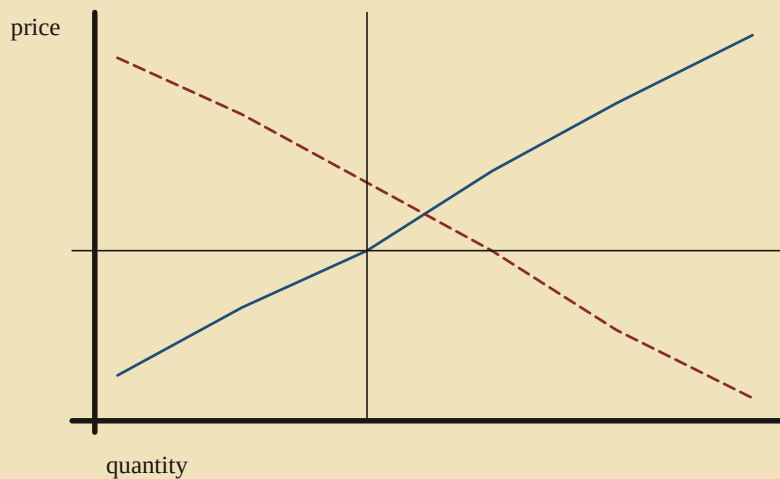


FIG. 11.4. a monopolistic competitor's short-run profit and long-run zero profit side by side

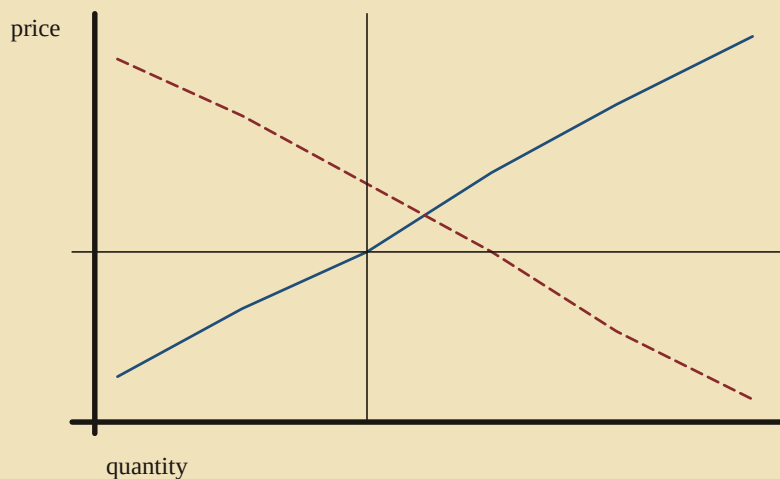


FIG. 11.5. a revenue comparison of a price cut for an elastic good and an inelastic good

WORKED CASE: THE RUNNING RECORD

Situation. Meghna and Sons (illustrative). Perfect competition: exercise books, market price Tk 60. Firm marginal cost by daily output: 10 books, Tk 40; 20, Tk 50; 30, Tk 60; 40, Tk 70; 50, Tk 80. Profit-maximising output 30 (marginal cost equals Tk 60). Average cost at 30 books Tk 52; profit $(60 - 52) \times 30 = \text{Tk } 240$ a day. Entry of new shops lowers price toward average cost in the long run. Monopoly: sole authorised seller of one foreign reference work. Price and weekly quantity: Tk 1,000, 10 copies; Tk 900, 20; Tk 800, 30; Tk 700, 40. Total revenue Tk 10,000; 18,000; 24,000; 28,000. Marginal revenue between steps Tk 800; 600; 400. Marginal cost constant Tk 500. Profit-maximising output 30 copies at Tk 800; profit $(800 - 500) \times 30 = \text{Tk } 9,000$. Monopolistic competition: a mystery novel series differentiated by cover and author; price Tk 400; short run average cost Tk 340 at 50 copies a week; profit Tk 3,000. After two more shops enter, the shop sells 40 copies at Tk 380 with average cost Tk 380; profit

The worked case applies the chapter without adding facts to the record.

nil. Pricing: price cut from Tk 60 to Tk 55 for the exercise book raised weekly sales from 200 to 210; revenue Tk 12,000 fell to Tk 11,550 (inelastic). Market structure ranking for the shop: exercise books near perfect competition; the sole-authorised reference work a monopoly; mystery novels monopolistic competition. Question. What does the record show when the chapter's central rule is applied to these facts?

The reasoning follows the data rather than an invented story.

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

The conclusion is limited to what the case supports.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

A final check asks whether the result fits the original data.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

The chapter has built one connected chain of reasoning.

Compare how firms behave when they face many rivals, one rival or none, and how each structure sets price and output. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

11.T1 find the profit-maximising output from a new cost table in each of three structures, name the structure from a list of features, decide whether a price cut helps revenue

11.T2 assess the view that firms always aim to maximise profit

WHERE PEOPLE GO WRONG

A common error is to apply a rule after its condition has changed.

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Another error is to confuse a numerical result with the meaning of that result.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The historical case shows the chapter idea in a documented setting.

The entry directs attention to The 1911 US Supreme Court decision to break up Standard Oil; search: Standard Oil 1911 Supreme Court dissolution. This account is used only for the historical connection specified by the chapter. It

illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

REVIEW QUESTIONS

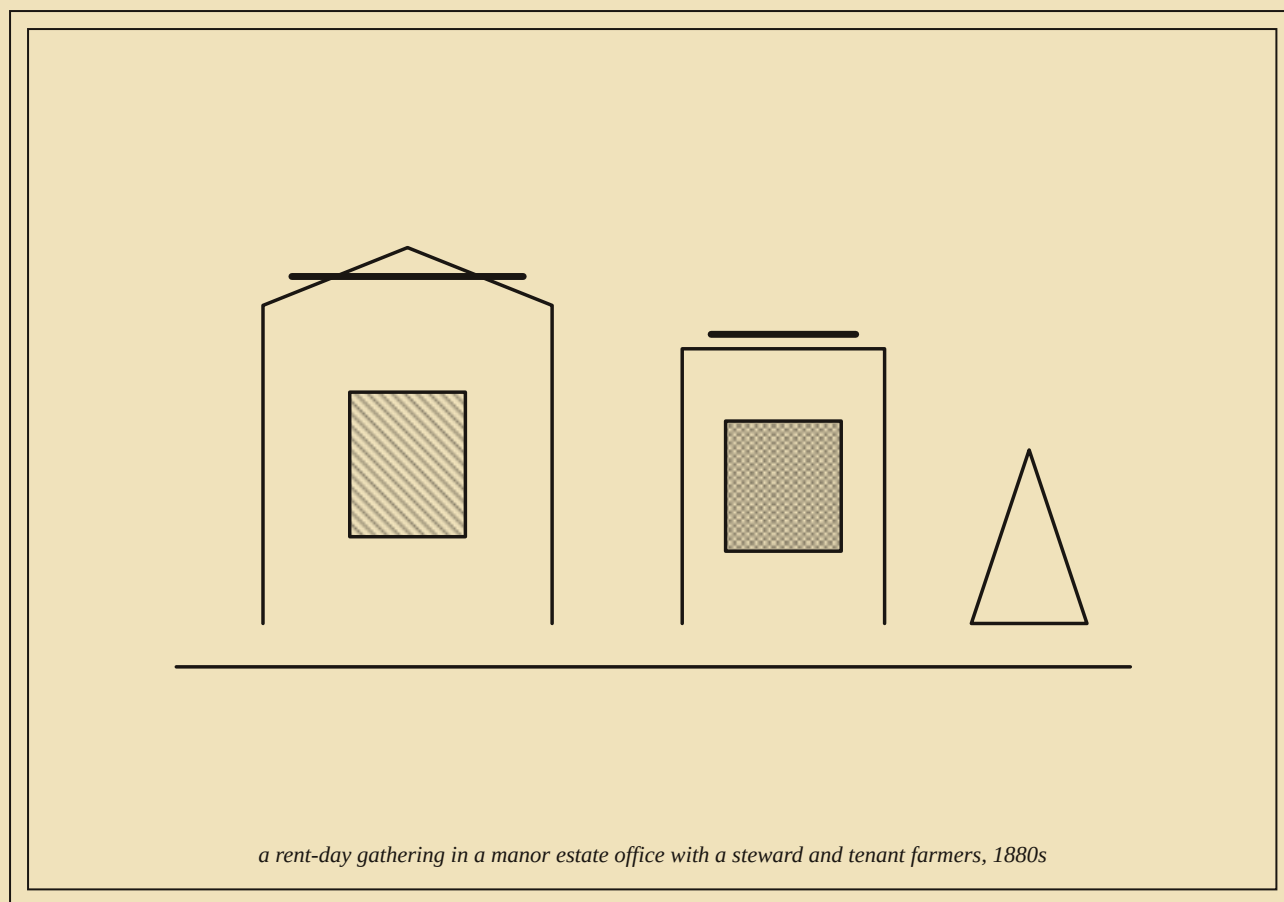
11.3 find the profit-maximising output from a new cost table in each of three structures, name the structure from a list of features, decide whether a price cut helps revenue

11.4 assess the view that firms always aim to maximise profit

11.5 after the price-taker passage and after the monopoly marginal-revenue table

PAUSE AND CHECK 11.P6 after the price-taker passage

PAUSE AND CHECK 11.P7 the monopoly marginal-revenue table



CHAPTER XII

Distribution: Rent, Wages, Interest and Profit

What does this chapter let us explain that the earlier ideas could not explain alone?



Explain how the money earned from production is shared among land, labour, capital and enterprise. The chapter builds the idea in small stages, then applies it to the running case. The examples keep the stated conditions visible so the reasoning can be checked.

The opening question sets up the problem before the rule is named.

The reader first separates the objects of the analysis.

The running case provides records rather than invented narrative.

Suppose you have to make a decision with a limited amount of money, time, space or equipment. Two choices may both be sensible, yet the same resources cannot serve both at once. The economic question is not whether one choice sounds attractive in isolation. It is what follows when the available resource is limited and a choice must be made. Explain how the money earned from production is shared among land, labour, capital and enterprise. This chapter develops that reasoning from a concrete setting to a general rule, then returns to the case so the rule can be checked.

Ask what is being chosen, who is making the choice, which quantity is being measured, and which conditions are being held fixed. This order prevents a common mistake: answering a different question from the one the evidence can answer. A price, a quantity, a cost, an input or a payment may appear in the same record, but each has a distinct role. The chapter therefore introduces each working term at its first necessary point and keeps its meaning stable afterwards.

Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000; nil. The shop itself pays Tk 60,000 a month in rent (carried base fact). Wages: shop assistant's nominal wage Tk 18,000 a month; general prices rose 8 per cent in the year; real wage about Tk 16,667 in last year's prices. Interest: a loan of Tk 200,000 at 11 per cent a year costs Tk 22,000 a year. Profit: yearly sales Tk 6,000,000; explicit costs Tk 5,400,000; accounting profit Tk 600,000. Implicit costs: owners' unpaid time Tk 360,000 and return on their own capital Tk 90,000, total Tk 450,000. Economic profit Tk 150,000; normal profit Tk 450,000. The case is illustrative, so it is used to show how the method works rather than to suggest facts beyond the record. Dates, amounts and decisions are treated as given data. When a conclusion depends on a condition, that condition is stated. When the case does not supply evidence for a claim, the discussion stays at the general level instead of creating a new fact.

* * *

12.1 | RENT | DEFINING THEORY | 225

This section asks what rent means before applying it.

Factor price. the payment a factor of production earns: rent for land, wages for labour, interest for capital, profit for enterprise

Factor price is the working idea for this part of the chapter.

Economic rent. the payment to a factor above the minimum needed to keep it in its present use

Economic rent is the working idea for this part of the chapter.

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Factor price** means the payment a factor of production earns: rent for land, wages for labour, interest for capital, profit for enterprise. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in rent, before moving to the next step.

The term **Economic rent** means the payment to a factor above the minimum needed to keep it in its present use. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before

rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in rent, before moving to the next step.

The term **Differential rent** means the extra return a better plot earns over the poorest plot in use. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in rent, before moving to the next step.

The term **Wage** means the payment made to labour for its work. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in rent, before moving to the next step.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

Before moving on, keep one question in view: what must remain true for the rule in rent to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

12.2 | WAGES | FRAMEWORK | 775

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Nominal wage** means the wage counted in money. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wages, before moving to the next step.

Differential rent. the extra return a better plot earns over the poorest plot in use

Differential rent is the working idea for this part of the chapter.

Wage. the payment made to labour for its work

Wage is the working idea for this part of the chapter.

Rent is understood by separating the question into small steps.

Rent is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what wages means before applying it.

Nominal wage. the wage counted in money

Nominal wage is the working idea for this part of the chapter.

Real wage. the amount of goods and services a nominal wage can buy

Real wage is the working idea for this part of the chapter.

Interest. the payment for the use of borrowed money or capital over time

Interest is the working idea for this part of the chapter.

Rate of interest. the interest paid in a year as a percentage of the amount borrowed

Rate of interest is the working idea for this part of the chapter.

Wages is understood by separating the question into small steps.

Wages is understood by separating the question into small steps.

Wages is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

This section asks what interest means before applying it.

The term **Real wage** means the amount of goods and services a nominal wage can buy. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wages, before moving to the next step.

The term **Interest** means the payment for the use of borrowed money or capital over time. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wages, before moving to the next step.

The term **Rate of interest** means the interest paid in a year as a percentage of the amount borrowed. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in wages, before moving to the next step.

The running case gives the idea a concrete setting without requiring a story or invented motives. The records identify dates, quantities and prices, so the reader can reproduce the reasoning. Where the case gives a number, the number is retained; where it gives a general rule, the explanation remains general rather than creating an unsupported shop record.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in wages to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

12.3 | INTEREST | FRAMEWORK | 225

The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the

reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Profit** means the reward of enterprise: what remains of revenue after the other factors are paid. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in interest, before moving to the next step.

The term **Economic profit** means revenue less all costs, explicit and implicit. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in interest, before moving to the next step.

The term **Accounting profit** means revenue less explicit costs only. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in interest, before moving to the next step.

The term **Normal profit** means the least profit that keeps a firm in business, equal to the implicit costs of the owner's own resources. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in interest, before moving to the next step.

The same reasoning can be transferred to another setting. A household, market stall, workshop or larger firm may have different numbers, but the analytical sequence remains: identify the decision, state the relevant condition, apply the rule, and inspect the result. This transfer is important because an economic idea is useful only when it can be recognised outside its first example.

There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

Before moving on, keep one question in view: what must remain true for the rule in interest to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

Profit. the reward of enterprise: what remains of revenue after the other factors are paid

Profit is the working idea for this part of the chapter.

Economic profit. revenue less all costs, explicit and implicit

Economic profit is the working idea for this part of the chapter.

Accounting profit. revenue less explicit costs only

Accounting profit is the working idea for this part of the chapter.

Normal profit. the least profit that keeps a firm in business, equal to the implicit costs of the owner's own resources

Normal profit is the working idea for this part of the chapter.

Interest is understood by separating the question into small steps.

Interest is understood by separating the question into small steps.

The section closes by restating the condition that controls the result.

12.4 | PROFIT | FRAMEWORK | 775

This section asks what profit means before applying it.

Risk bearing. taking on the chance of loss when production is uncertain

Risk bearing is the working idea for this part of the chapter.

Marginal productivity theory. the view that each factor is paid in line with the extra output its last unit adds

Marginal productivity theory is the working idea for this part of the chapter.

Profit is understood by separating the question into small steps.

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The section is organised around a question that can be answered from the definitions, the case record and the stated conditions. Start with the smallest version of the problem. Then add the detail that changes the result. This order makes the reasoning visible and reduces the temptation to memorise a formula without understanding what its parts represent.

The term **Risk bearing** means taking on the chance of loss when production is uncertain. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in profit, before moving to the next step.

The term **Marginal productivity theory** means the view that each factor is paid in line with the extra output its last unit adds. For the running case, the record gives a concrete numerical or organisational setting in which this idea can be seen. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000. In this chapter, the term matters because it lets the reader state precisely what is being measured or compared in profit, before moving to the next step.

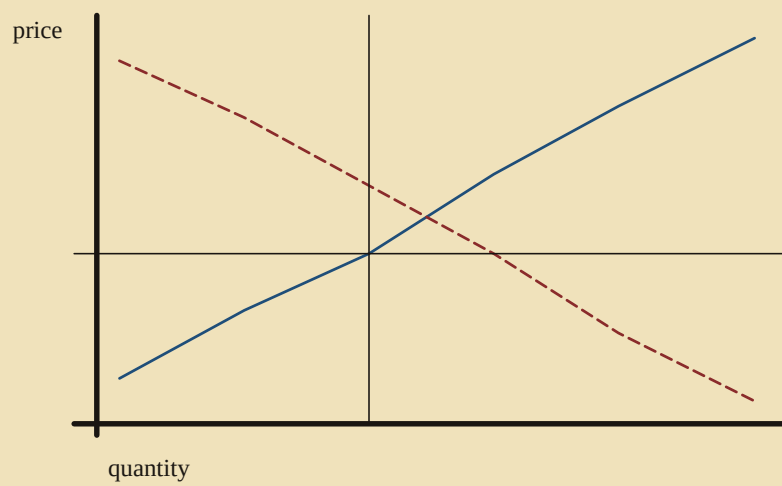
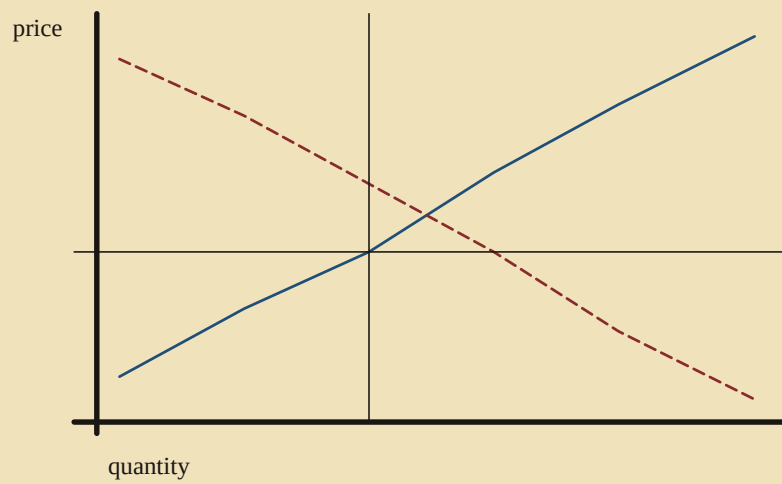
There is also a limit to the rule. Economic relationships describe tendencies under stated conditions, not mechanical laws that ignore every other fact. If a condition changes, the result may change as well. A careful reader therefore states the condition before applying the rule and checks whether the new situation still satisfies it.

The central question in profit is how a decision follows from limited information and a stated rule. A useful first step is to name the objects being compared, then keep the conditions fixed while one relationship is examined. This prevents several different causes from being mixed together. The result is a description that can be checked against the figures or the stated situation.

A beginner can read this idea in two stages. First, ask what changes and what stays unchanged. Next, ask what the change does to the quantity, cost, price, satisfaction or output under discussion. The distinction is small in wording but large in analysis, because the same numerical change can have different meanings when the underlying condition is different.

Before moving on, keep one question in view: what must remain true for the rule in profit to apply? The answer is contained in the definitions and examples above. If the condition changes, do not reuse the conclusion automatically. Re-state the problem, identify the new condition, and then apply the rule again.

* * *

FIG. 12.1. *differential rent on three plots of different yield*FIG. 12.2. *a labour market with demand and supply setting the wage*

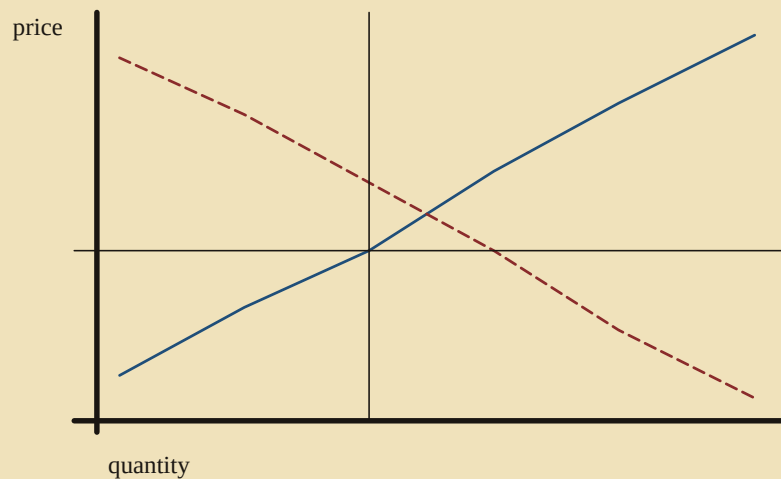


FIG. 12.3. a loanable-funds diagram setting the rate of interest

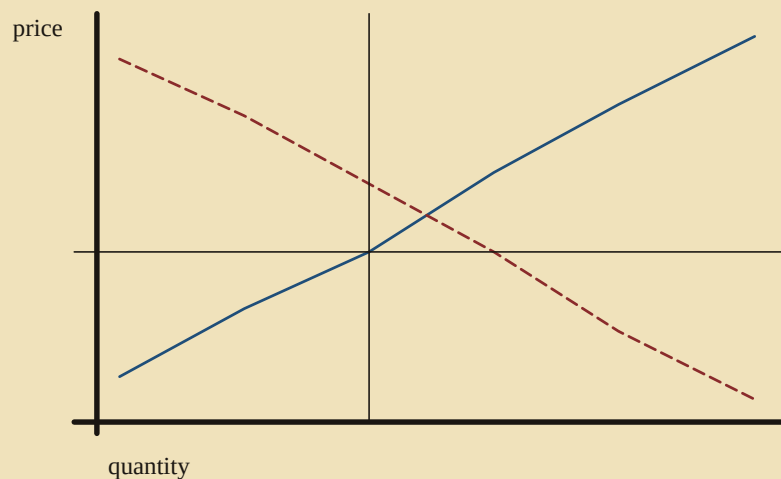


FIG. 12.4. a stacked bar splitting the shop's yearly sales into costs, normal profit and economic profit

WORKED CASE: THE RUNNING RECORD

The worked case applies the chapter without adding facts to the record.

Situation. Meghna and Sons (illustrative). Rent: three locations for a stall, yearly figures before rent: main road Tk 150,000; side lane Tk 90,000; marginal plot Tk 60,000 (pays no rent). Differential rent: Tk 90,000; Tk 30,000; nil. The shop itself pays Tk 60,000 a month in rent (carried base fact). Wages: shop assistant's nominal wage Tk 18,000 a month; general prices rose 8 per cent in the year; real wage about Tk 16,667 in last year's prices. Interest: a loan of Tk 200,000 at 11 per cent a year costs Tk 22,000 a year. Profit: yearly sales Tk 6,000,000; explicit costs Tk 5,400,000; accounting profit Tk 600,000. Implicit costs: owners' unpaid time Tk 360,000 and return on their own capital Tk 90,000, total Tk 450,000. Economic profit Tk 150,000; normal profit Tk 450,000. Question. What does the record show when the chapter's central rule is applied to these facts?

The reasoning follows the data rather than an invented story.

Reasoning. Begin by listing the given figures and identifying the relationship that the chapter explains. Where arithmetic is needed, write the starting figures

first, then the change, then the result. Where the issue is descriptive, name the category and give the condition that places the case in that category. This keeps the method reproducible: another reader should be able to follow the same sequence and reach the same conclusion from the stated record.

Conclusion. The record illustrates the chapter's central relationship in a controlled setting. It does not prove that the same result must occur in every market, firm or household. The value of the worked case is that it shows how a general idea is applied while the assumptions remain visible.

Check. Re-read the quantities, prices, costs or inputs used in the calculation. Confirm that units and periods match and that no figure has been silently changed. If the result contradicts a stated fact, return to the earlier step rather than adjusting the conclusion to fit the expected answer.

* * *

SUMMARY

Explain how the money earned from production is shared among land, labour, capital and enterprise. The key terms now have fixed meanings, the running case has shown their use, and the limits of the rules have been kept visible. The next chapter can therefore treat these ideas as working tools rather than introduce them again from the beginning.

THINK IT THROUGH

12.T1 compute differential rents for new plots, compute a real wage from new price data, separate economic profit from accounting profit in a new account

12.T2 explain why the poorest plot in use earns no rent

WHERE PEOPLE GO WRONG

Correction. State the condition first. Then decide whether the new situation still satisfies it. If it does, the rule can be applied again; if it does not, the analysis must be adjusted.

Correction. A number becomes useful only after its unit, period and economic meaning are identified. Always say what the result measures and which figures produced it.

IN HISTORY

The entry directs attention to David Ricardo's rent theory in his 1815 essay and his 1817 *Principles*, and the Corn Laws repealed in 1846; search: Ricardo rent Corn Laws 1815 repeal 1846. This account is used only for the historical connection specified by the chapter. It illustrates why the economic problem mattered in practice, while the chapter's analytical rules remain separate from the historical narrative.

SOURCES

Further reading and historical attribution should be checked against the original works and reliable archival or institutional sources. The chapter does not require the reader to

The conclusion is limited to what the case supports.

A final check asks whether the result fits the original data.

The chapter has built one connected chain of reasoning.

A common error is to apply a rule after its condition has changed.

Another error is to confuse a numerical result with the meaning of that result.

The historical case shows the chapter idea in a documented setting.

The chapter uses a small number of historical references to anchor its defining ideas.

accept a historical interpretation merely because it appears in a secondary account.

REVIEW QUESTIONS

12.3 compute differential rents for new plots, compute a real wage from new price data, separate economic profit from accounting profit in a new account

12.4 explain why the poorest plot in use earns no rent

12.5 after the differential-rent passage and after the economic-versus-accounting profit passage

PAUSE AND CHECK 12.P6 after the differential-rent passage

PAUSE AND CHECK 12.P7 the economic-versus-accounting profit passage

APPENDIX A

Maths You Will Use



This reference keeps the arithmetic used later in one place.

The reader can return here whenever a calculation needs a percentage, ratio, average, rearranged equation or graph reading. Each method starts with the smallest useful example and then moves to a slightly less tidy one.

* * *

A.1 A1.1 | PERCENTAGES AND PERCENTAGE CHANGE | TECHNIQUE | 250

Percentages and percentage change is a tool for checking numerical relationships.

In percentages and percentage change, write the given numbers before choosing an operation. Keep the units visible, carry the arithmetic line by line, and round only at the point required by the question. For a percentage change, compare the change with the starting value. For an average, add the values and divide by their count. For an equation, perform the same operation on both sides. For a graph, read the axes before reading a point.

Given a price of Tk 60 rising to Tk 66, the change is Tk 6. The percentage change is $6 \div 60 \times 100 = 10$ per cent. The important check is that the starting value is 60, not 66. The same discipline applies to ratios, equations and graphs: identify the reference quantity before calculating.

* * *

A.2 A1.2 | RATIOS, AVERAGES AND ROUNDING | TECHNIQUE | 250

Ratios, averages and rounding is a tool for checking numerical relationships.

In ratios, averages and rounding, write the given numbers before choosing an operation. Keep the units visible, carry the arithmetic line by line, and round only at the point required by the question. For a percentage change, compare the change with the starting value. For an average, add the values and divide by their count. For an equation, perform the same operation on both sides. For a graph, read the axes before reading a point.

Given a price of Tk 60 rising to Tk 66, the change is Tk 6. The percentage change is $6 \div 60 \times 100 = 10$ per cent. The important check is that the starting value is 60, not 66. The same discipline applies to ratios, equations and graphs: identify the reference quantity before calculating.

* * *

A.3 A1.3 | REARRANGING AN EQUATION | TECHNIQUE | 250

Rearranging an equation is a tool for checking numerical relationships.

In rearranging an equation, write the given numbers before choosing an operation. Keep the units visible, carry the arithmetic line by line, and round only at the point required by the question. For a percentage change, compare the change with the starting value. For an average, add the values and divide by their count. For an equation, perform the same operation on both sides. For a graph, read the axes before reading a point.

Given a price of Tk 60 rising to Tk 66, the change is Tk 6. The percentage change is $6 \div 60 \times 100 = 10$ per cent. The important check is that the starting value is 60, not 66. The same discipline applies to ratios, equations and graphs: identify the reference quantity before calculating.

* * *

A tiny worked example makes the procedure concrete.

A.4 A1.4 | READING A GRAPH | TECHNIQUE | 250

In reading a graph, write the given numbers before choosing an operation. Keep the units visible, carry the arithmetic line by line, and round only at the point required by the question. For a percentage change, compare the change with the starting value. For an average, add the values and divide by their count. For an equation, perform the same operation on both sides. For a graph, read the axes before reading a point.

Given a price of Tk 60 rising to Tk 66, the change is Tk 6. The percentage change is $6 \div 60 \times 100 = 10$ per cent. The important check is that the starting value is 60, not 66. The same discipline applies to ratios, equations and graphs: identify the reference quantity before calculating.

Reading a graph is a tool for checking numerical relationships.

A tiny worked example makes the procedure concrete.

GLOSSARY

Accounting profit. revenue less explicit costs only. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Art. the skilled use of knowledge to reach a practical aim. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Average cost. total cost divided by the number of units produced. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Average product. total product divided by the number of units of the variable input. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Backward-bending supply curve. a curve showing that beyond some pay level a worker offers fewer hours as pay rises further. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Bandwagon effect. a rise in demand for a good because more people are seen to buy it. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Bilateral monopoly. a market with a single seller facing a single buyer. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Budget line. a line showing every bundle of two goods a consumer can exactly afford with a given income. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Cardinal measurement. measuring satisfaction in numbers, such as 20 utils, so amounts can be added and compared. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Ceteris paribus. a Latin phrase meaning other things being equal, used to study one change at a time. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Choice. picking one option and giving up the others because resources cannot cover every option. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Competitive market. a market with many buyers and sellers, so no single one can set the price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Complementary goods. goods used together, so a rise in one's price lowers demand for the other. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Constant returns to scale. output rises by the same proportion as inputs. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Consumer. a person who buys or uses goods and services to meet wants. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Consumer equilibrium. the affordable bundle that gives a consumer the greatest satisfaction. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Consumer surplus. the gap between what a buyer would pay and what is actually paid. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Consumption. the using up of goods and services to meet wants. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Cost. the value of the resources used up in making a good or service. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Cross elasticity of demand. the percentage change in quantity demanded of one good divided by the percentage change in another good's price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Decreasing returns to scale. output rises by a smaller proportion than inputs. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Demand. the quantity of a good people are willing and able to buy at each possible price, in a given period. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Demand curve. a line on a graph showing the quantity demanded at each price, other things equal. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Demand schedule. a table listing the quantity demanded at each of several prices. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Determinants of demand. the things other than the good's own price that change how much people want to buy. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Determinants of supply. the things other than the good's own price that change how much sellers offer. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Differential rent. the extra return a better plot earns over the poorest plot in use. The chapter uses this definition consistently; the running example

shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Economic law. a statement of a regular tendency in human choices, true when other things stay equal. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Economic organisation. the way a society or business arranges who decides what is made, how and for whom. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Economic profit. revenue less all costs, explicit and implicit. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Economic rent. the payment to a factor above the minimum needed to keep it in its present use. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Economic system. the set of rules and institutions a country uses to answer its basic economic problems. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Economics. the study of how people, firms and societies choose among uses of limited resources to meet wants. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Elastic demand. demand for which the percentage fall in quantity is greater than the percentage rise in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Elastic supply. supply for which quantity responds by a larger percentage than price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Elasticity. a measure of how strongly one quantity responds when another quantity changes. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Equilibrium. a state in which nothing pushes the price or quantity to change. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Equilibrium price. the price at which quantity demanded equals quantity supplied. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Equilibrium quantity. the quantity bought and sold at the equilibrium price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Excess demand. the amount by which quantity demanded exceeds quantity supplied at a price below equilibrium. The chapter uses this definition consistently; the running example

shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Excess supply. the amount by which quantity supplied exceeds quantity demanded at a price above equilibrium. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Explicit cost. a cost paid out in money to others, such as rent or wages. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Factor of production. one of the four kinds of resource used in making things: land, labour, capital, enterprise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Factor price. the payment a factor of production earns: rent for land, wages for labour, interest for capital, profit for enterprise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Firm. an organisation that uses resources to make goods or services and sell them. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Fixed cost. a cost that stays the same however much is produced in the short run. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Fixed input. an input whose quantity cannot be changed in the short run. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Giffen good. a rare inferior good for which a price rise leads buyers to buy more of it. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Good. a physical thing people buy to satisfy a want, such as a book or a pen. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Household. a person or group living together that buys goods and services and supplies resources. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Implicit cost. the value of resources the owner supplies, such as own time, for which no payment is made. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Income elasticity of demand. the percentage change in quantity demanded divided by the percentage change in buyers' income. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Increasing returns to scale. output rises by a larger proportion than inputs. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Indifference curve. a line joining bundles of two goods that give a consumer the same satisfaction. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Indifference map. a set of indifference curves, with curves farther from the origin showing more satisfaction. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Industry. all the firms that make the same or very similar goods. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Inelastic demand. demand for which the percentage fall in quantity is smaller than the percentage rise in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Inelastic supply. supply for which quantity responds by a smaller percentage than price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Inferior good. a good for which demand falls when buyers' incomes rise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Input. a resource used in production. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Interest. the payment for the use of borrowed money or capital over time. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Isocost line. a line joining the input combinations that all cost the same total. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Isoquant. a curve joining the input combinations that all make the same output. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Law of demand. when the price of a good rises, quantity demanded falls, and when it falls, quantity demanded rises, other things equal. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Law of diminishing marginal utility. as a person consumes more units of a good, the extra satisfaction from each further unit falls. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Law of diminishing returns. when more of one input is added to fixed inputs, the extra output eventually falls. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Law of supply. when the price of a good rises, quantity supplied rises, and when it falls, quantity supplied falls, other things equal. The chapter uses this

definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Least-cost combination. the mix of inputs that makes a given output for the lowest total cost. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Long run. a period long enough to change every input. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Long-run average cost curve. a curve showing the lowest average cost of each output when every input can be changed. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Macroeconomics. the study of the economy as a whole, such as total output, jobs and the general price level. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Marginal cost. the extra cost of producing one more unit. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Marginal product. the extra output from adding one more unit of the variable input. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Marginal productivity theory. the view that each factor is paid in line with the extra output its last unit adds. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Marginal rate of substitution. the amount of one good a consumer will give up to gain one more unit of the other. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

Marginal revenue. the extra revenue gained by selling one more unit. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Marginal utility. the extra satisfaction gained from consuming one more unit of a good. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Market. any arrangement that brings buyers and sellers of a good together. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Market demand. the sum of all individual buyers' demands for a good at each price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Market structure. the features of a market, such as number of sellers, type of product and ease of entry, that shape how firms behave. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Microeconomics. the study of choices made by single households, firms and markets. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Model. a simplified picture of a situation that keeps only the features needed to answer a question. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Monopolistic competition. a market with many sellers of similar but not identical goods and free entry. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Monopoly. a market with one seller and no close substitute, so that seller sets the price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Nominal wage. the wage counted in money. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Normal good. a good for which demand rises when buyers' incomes rise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Normal profit. the least profit that keeps a firm in business, equal to the implicit costs of the owner's own resources. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Normative statement. a claim about what ought to be, resting on values and opinion. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Oligopoly. a market with a few large sellers, each aware of the others' actions. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Opportunity cost. the value of the best alternative given up when a choice is made. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Ordinal measurement. ranking satisfaction as first, second and third without giving it a number. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Output. the goods or services produced. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Percentage change. the size of a change divided by the starting value, multiplied by 100. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Perfect competition. a market with many sellers, an identical product, free entry and full information, so each firm takes the price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Perfectly elastic demand. demand that vanishes completely for any price rise, however small, shown as a flat curve. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Perfectly inelastic demand. demand that does not change at all when price changes, shown as a vertical curve. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Positive statement. a claim about what is, which can be checked against facts. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Price ceiling. a legal maximum price that may be charged for a good. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Price elasticity of demand. the percentage change in quantity demanded divided by the percentage change in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Price elasticity of supply. the percentage change in quantity supplied divided by the percentage change in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Price floor. a legal minimum price that may be charged for a good. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Price maker. a firm large enough that its own output choices affect the price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Price taker. a firm that must accept the market price because it is too small to change it. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Prime cost. the direct cost of materials and labour that go into a unit of output. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Producer surplus. the gap between the price a seller receives and the lowest price the seller would accept. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Product differentiation. making a firm's good different from rivals' in quality, design, brand or service. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Production. the process of turning resources into goods and services. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Production function. a rule or table linking the quantities of inputs used to the greatest output they can make. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Profit. the reward of enterprise: what remains of revenue after the other factors are paid. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Profit maximisation. choosing the output at which the gap between total revenue and total cost is greatest. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Pure competition. a market with many sellers, an identical product and free entry, where buyers may lack full information or free movement. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Pure monopoly. a monopoly in which one seller supplies the whole market with a product that has no close substitute. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XI*)

Quantity demanded. the amount people plan to buy at one particular price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Quantity supplied. the amount sellers plan to offer at one particular price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Rate of interest. the interest paid in a year as a percentage of the amount borrowed. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Real wage. the amount of goods and services a nominal wage can buy. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Resource. anything used to make goods and services: land, labour, capital and enterprise. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Returns to scale. the change in output when all inputs are changed by the same proportion. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Risk bearing. taking on the chance of loss when production is uncertain. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Saturation point. the quantity at which marginal utility reaches zero and total utility stops rising. The chapter uses this definition consistently; the running

example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Scarcity. the condition in which resources are too few to satisfy all the wants people have. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Science. a body of knowledge built by observing facts, forming explanations and testing them. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter II*)

Service. an activity done for someone, such as binding a thesis or delivering parcels. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Short run. a period too short to change at least one input, such as machines or floor space. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Snob effect. a fall in demand for a good because more people begin to own it. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Substitute goods. goods that can replace each other, so a rise in one's price raises demand for the other. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IV*)

Supplementary cost. the cost of items apart from prime cost, such as rent and equipment, which remain even when output is nil. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Supply. the quantity of a good sellers are willing and able to offer at each possible price, in a given period. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Supply curve. a line on a graph showing the quantity supplied at each price, other things equal. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Supply schedule. a table listing the quantity supplied at each of several prices. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VII*)

Total cost. fixed cost plus variable cost at a given output. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Total product. the whole output from a given amount of inputs. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Total revenue. the money a seller receives from sales, found by price multiplied by quantity sold. The chapter uses this definition consistently; the running example

shows how the term works before it is applied to a changed situation. (*Chapter V*)

Total surplus. consumer surplus plus producer surplus, a measure of the gain from trade. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VIII*)

Total utility. the whole satisfaction from all the units of a good consumed in a period. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Unit elastic demand. demand for which the percentage change in quantity equals the percentage change in price. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter V*)

Util. an imaginary unit used to count satisfaction in examples. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Utility. the satisfaction a person gets from using a good or service. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter III*)

Variable cost. a cost that rises and falls with the amount produced. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter X*)

Variable input. an input whose quantity can be changed even in the short run. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter IX*)

Wage. the payment made to labour for its work. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter XII*)

Want. something a person would like to have, whether or not they can pay for it. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter I*)

Willingness to pay. the highest price a buyer would accept to get one unit of a good. The chapter uses this definition consistently; the running example shows how the term works before it is applied to a changed situation. (*Chapter VI*)

ANSWERS TO QUESTIONS

CHAPTER I · SCARCITY, CHOICE AND THE ECONOMIC PROBLEM

1.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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1.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

1.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER II · ECONOMIC REASONING, LAWS AND THE SCOPE OF MICROECONOMICS

2.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

2.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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2.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

2.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

2.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER III · UTILITY AND DIMINISHING MARGINAL UTILITY

3.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

3.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

3.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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3.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

3.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER IV · DEMAND AND THE DEMAND CURVE

4.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

4.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

4.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

4.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

4.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

4.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

4.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER V · ELASTICITY OF DEMAND

5.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

5.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER VI · CONSUMER CHOICE: SURPLUS AND INDIFFERENCE CURVES

6.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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6.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

6.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER VII · SUPPLY AND ELASTICITY OF SUPPLY

7.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

7.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER VIII · HOW MARKETS FIND A PRICE

8.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

8.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

8.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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8.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

8.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

8.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER IX · PRODUCTION

9.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.4 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

9.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER X · COSTS OF PRODUCTION

10.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

10.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

10.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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10.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

10.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER XI · MARKET STRUCTURES AND THE FIRM

11.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

11.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

11.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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11.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

11.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

11.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

CHAPTER XII · DISTRIBUTION: RENT, WAGES, INTEREST AND PROFIT

12.t1 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

12.t2 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

12.3 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

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12.5 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

12.p6 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

12.p7 Apply the definition and conditions stated in the chapter. For a calculation, show the relevant change and divide or multiply only after identifying the correct starting value and unit.

FURTHER READING

Alfred Marshall introduces elasticity of demand in *Principles of Economics*, 1890, historical case reference

David Ricardo's rent theory in his 1815 essay and his 1817 *Principles*, and the Corn Laws repealed in 1846, historical case reference

Jules Dupuit, 1844, on measuring the usefulness of public works such as bridge tolls, historical case reference

The marginal revolution of 1871 to 1874: Jevons, Menger and Walras explain the diamond-and-water puzzle, historical case reference

US petrol shortage after the October 1973 oil embargo, with queues and the odd-even plate rationing of 1974, historical case reference

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